

8

Standard Costs and Variances

Figure 8.1 Coffee Shop. How do they know what each cup of coffee, muffin, or bagel costs so they can determine what price to charge? (credit: modification of "Bakery Coffee" by "veerasantinithi"/Pixabay, CC0)

Chapter Outline

LO **8.1** Explain How and Why a Standard Cost Is Developed
LO **8.2** Compute and Evaluate Materials Variances
LO **8.3** Compute and Evaluate Labor Variances
LO **8.4** Compute and Evaluate Overhead Variances
LO **8.5** Describe How Companies Use Variance Analysis

Why It Matters

Sam saw how much coffee his fellow students were drinking and decided to open a student-run coffee shop on campus. Sam knew that developing a plan for the coffee shop would help make the shop successful. He researched what types of coffee to offer, the hours the shop would be open, and the number of employees needed, by researching other coffee shops near campus. He brewed coffee to determine the cost of the coffee and the time it took to brew. He also served several friends to determine how long it would take to serve customers. He observed, in other coffee shops, how much cream and other additives are used by customers. He talked to several coffee suppliers for prices of his various materials. He looked at empty stores near campus to determine what his rent would be. Now that he has this information, he is not sure how to make it useful to him. How could he use this information to plan and control the operations of the shop? One calculation he can do is determine his standard costs.

What is the difference between a *budget* and a *standard*? A budget usually refers to a company's projections for costs, revenues, and cash flows associated with the overall operations of the organization, or a subsection of the corporation such as a division. A standard usually refers to a company's projected costs for a single unit of a product or service and includes the expected, or standard, cost for the various cost components of each unit, such as materials, labor, and overhead.

8.1 Explain How and Why a Standard Cost Is Developed

A syllabus is one way an instructor can communicate expectations to students. Students can use the syllabus to plan their studying to maximize their grade and to coordinate the amount and timing of studying for each course. Knowing what is expected, and when it is expected, allows for better plans and performance. When your performance does not match your expectations, a **variance** arises—a difference between the standard and the actual performance. You then need to determine why the difference occurred. You want to know why you did not receive the grade you expected so you can make adjustments for the next assignment to earn a better grade.

Companies operate in a similar manner. They have an expectation, or **standard**, for production. For example, if a company is producing tables, it might establish standards for such components as the amount of board feet of lumber expected to be used in producing each table or the number of hours of direct labor hours it expected to use in the table's production. These standards can then be used in establishing standard costs that can be used in creating an assortment of different types of budgets.

When a variance occurs in its standards, the company investigates to determine the causes, so they can perform better in the future. For example, General Motors has standards for each item on a vehicle. It can determine the cost and selling price of a power antenna by knowing the standard material cost for the antenna and the standard labor cost of adding the antenna to the vehicle. General Motors also can add up all of the standard times for all vehicles it makes to determine if too much or too little labor was used in production.

LINK TO LEARNING

Developing standards is a complicated and costly process. Review this article on how to develop a standard cost system (https://openstax.org/l/50StandardCost) for more details.

Fundamentals of Standard Costs

It is important to establish standards for cost at the beginning of a period to prepare the budget; manage material, labor, and overhead costs; and create a reasonable sales price for a good. A **standard cost** is an expected cost that a company usually establishes at the beginning of a fiscal year for prices paid and amounts used. The standard cost is an expected amount paid for materials costs or labor rates. The standard quantity is the expected usage amount of materials or labor. A standard cost may be determined by past history or industry norms. The company can then compare the standard costs against its actual results to measure its efficiency. Sometimes when comparing standard costs against actual results, there is a difference.

This difference can be attributed to many reasons. For example, the coffee company mentioned in the opening vignette may expect to pay $0.50 per ounce for coffee grounds. After the company purchased the coffee grounds, it discovered it paid $0.60 per ounce. This variance would need to be accounted for, and possible operational changes would occur as a result. Cost accounting systems become more useful to management when they include budgeted amounts to serve as a point of comparison with actual results.

Many departments help determine standard cost. Product design, in conjunction with production, purchasing, and sales, determines what the product will look like and what materials will be used. Production works with purchasing to determine what material will work best in production and will be the most cost efficient. Sales will also help decide the material in terms of customer demand. Production will work with personnel to determine labor costs for the product, which is based on how long it will take to make the product, which departments will be involved, and what type and number of employees it will take.

Consider how many different materials can go into a product. For example, there are approximately 14,000 parts that comprise the average automobile. The manufacturer will set a standard price and a quantity used per automobile for each part, and it will determine the labor required to install the part. At Fiat Chrysler Automobiles' Belvidere Assembly Plant, for example, there are approximately 5,000 employees assembling automobiles.[1] In addition to having standard costs associated with each part, each employee has standards for the job he or she performs.

Standard costs are typically established for reasonably attainable levels of efficiency (production). They serve as a target and are useful in motivating standard performance. An **ideal standard** level is set assuming that everything is perfect, machines do not break down, employees show up on time, there are no defects, there is no scrap, and materials are perfect. This level of standard is not the best motivator, because employees may see this level as unattainable. For example, consider whether you would take a course if the letter grades were as follows: an A is 99–100%, a B is 98–99%, a C is 97–98%, a D is 96–97%, and below 96% is an F. These standards are unreasonable and unrealistic, and they would not motivate students to do well in the course.

At the other end of the spectrum, if the standards are too easy, there is little motivation to do better, and products may not be properly built, may be built with inferior materials, or both. For example, consider how you would handle the following grading scale for your course: an A is 50–100%, a B is 35–50%, a C is 10–35%, a D is 2–10%, and below 2% is an F. Would you learn anything? Would you try very hard? The same considerations come into play for employees with standards that are too easy.

Instead of these two extremes, a company would set an **attainable standard**, which is one that employees can reach with reasonable effort. The standards are not so high that employees will not try to reach them and not so low that they do not give any incentive for employees to achieve profitability.

In order for a company to establish its attainable standard cost for each product, it must consider the standard costs for materials, labor, and overhead. The material standard cost consists of a standard price per unit of material and a standard amount of material per unit. Returning to the opening vignette, let us say the coffee shop is trying to establish the standard materials cost for one cup of regular coffee. To keep the example simple, we are not incorporating the cost of water or the ceramic cup cost (since they are reused). Two components for the cup of coffee will need to be considered:

1. Price per ounce of coffee grounds
2. Amount of coffee grounds (materials) used per cup of coffee

To determine the standards for labor, the coffee shop would need to consider two additional components:

1. Labor rate per minute
2. Amount of time to make one cup of regular coffee

To determine the standard for overhead, the coffee shop would first need to consider the fact that it has two types of overhead as shown in Figure 8.2. Greater detail about the calculation of the variable and fixed overhead is provided in Compute and Evaluate Overhead Variances.

1. Fixed overhead (does not change in total with production)
2. Variable overhead (does change in total with production)

All of this information is entered on a standard cost card.

1 "Belvidere Assembly Plant and Belvidere Satellite Stamping Plant." Fiat Chrysler Automobiles. June 2018. http://media.fcanorthamerica.com/newsrelease.do?id=323&mid=1

STANDARD COST CARD Product: 1 Cup of Coffee				
Manufacturing Cost Information	Standard Quantity	×	Standard Cost per Unit	= Cost Summary
Direct Materials				
Material (coffee grounds)	0.5 ounces		$.50 per ounce	$0.25
Direct Labor				
Barista	1 minute		$0.20 per minute	$0.20
Manufacturing Overhead				
Variable overhead	1 minute		$0.05 per direct labor minute	$0.05
Fixed overhead	1 minute		$0.10 per direct labor minute	$0.10
Standard Cost per cup of coffee				**$0.60**

Figure 8.2 Standard Cost Card for a Coffee Shop. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

Once a company determines a standard cost, they can then evaluate any variances. A variance is the difference between a standard cost and actual performance. There are favorable and unfavorable variances. A **favorable variance** involves spending less, or using less, than the anticipated or estimated standard. An **unfavorable variance** involves spending more, or using more, than the anticipated or estimated standard. Before determining whether the variance is favorable or unfavorable, it is often helpful for the company to determine why the variance exists.

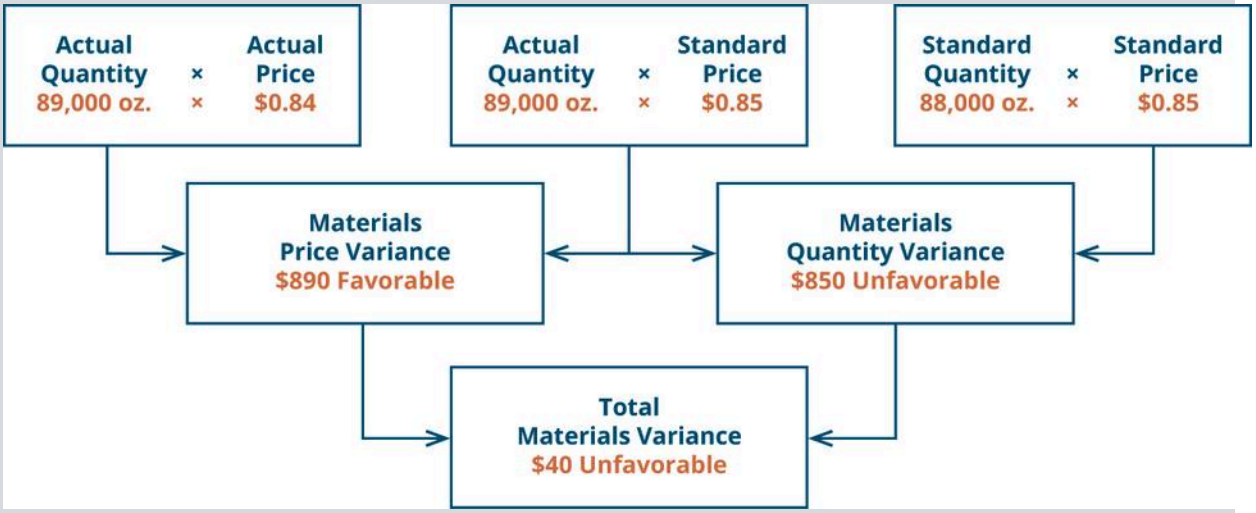
YOUR TURN

Developing a Standard Cost Card

Use the information provided to create a standard cost card for production of one deluxe bicycle from Bicycles Unlimited.

To make one bicycle it takes four pounds of material. The material can usually be purchased for $5.25 per pound. The labor necessary to build a bicycle consists of two types. The first type of labor is assembly, which takes 2.75 hours. These workers are paid $11.00 per hour. The second type of labor is finishing, which takes 4 hours. These workers are paid $15.00 per hour. Overhead is applied using labor hours. The variable overhead rate is $5.00 per labor hour. The fixed overhead rate is $3.00 per hour.

Solution



ETHICAL CONSIDERATIONS

Ethical Variance Analysis

Variance analysis allows managers to see whether costs are different than planned. Once a difference between expected and actual costs is identified, variance analysis should delve into why the costs differ and what the magnitude of the difference means. To determine why a cost differs, it should be established if the additional cost provides a benefit or detriment to an organization's stakeholders, the people or entities that are affected by the organization's actions or inactions. Not all stakeholders are equal in the analysis, but an organization should recognize each stakeholder's interest in the organization's business and operational decisions, while ranking the importance of the stakeholder in relation to any decision made.

Ranking should look to how stakeholders are affected by costs and any decisions related to cost variance, or why the variance occurred. For example, if a cost variance is due to an additional cost to make a product eco-friendly, then an organization may determine that incurring the cost is a benefit to its stakeholders. However, if the additional cost creates an unfavorable situation for a stakeholder, the process incurring the cost should be investigated. Remember that the owners of a company, including shareholders, are also stakeholders. To determine the best course of action for an organization, cost analysis should help inform stakeholder analysis—the process of systematically gathering and analyzing all of the information related to a business decision.

Different factors may produce a variance. The company could have paid too much or too little for production. It may have purchased the wrong grade of material or hired employees with more or less experience than required. Sometimes the variances are interrelated. For example, purchasing substandard materials may lead to using more time to make the product and may produce more scrap. The substandard material may have been more difficult to work with or had more defects than the proper grade material. In such a situation, a favorable material price variance could cause an unfavorable labor efficiency variance and an unfavorable material quantity variance. Employees who do not have the expected experience level may save money in the wage rate but may require more hours to be worked and more material to be used because of their inexperience.

Another situation in which a variance may occur is when the cost of labor and/or material changes after the standard was established. Toward the end of the fiscal year, standards often become less reliable because time has passed and the environment has changed. It is not reasonable to expect the price of all materials and labor to remain constant for 12 months. For example, the grade of material used to establish the standard may no longer be available.

Manufacturing Cost Variances

As you've learned, the standard price and standard quantity are anticipated amounts. Any change from these budgeted amounts will produce a variance. There can be variances for materials, labor, and overhead. Direct materials may have a variance in price of materials or quantity of materials used. Direct labor may have a variance in the rate paid to workers or the amount of time used to make a product. Overhead may produce a variance in expected fixed or variable costs, leading to possible differences in production capacity and management's ability to control overhead. More specifics on the formulas, processes, and interpretations of the direct materials, direct labor, and overhead variances are discussed in each of this chapter's following sections.

CONCEPTS IN PRACTICE

Qualcomm[2]

Qualcomm Inc. is a large producer of telecommunications equipment focusing mainly on wireless products and services. As with any company, Qualcomm sets labor standards and must address any variances in labor costs to stay on budget, and control overall manufacturing costs.

In 2018, Qualcomm announced a reduction to its labor force, affecting many of its full-time and temporary workers. The reduction in labor was necessary to suppress rising expenses that could not be controlled through overhead or materials cost-cutting measures. The variances between standard labor rates and actual labor rates, and diminishing profit margins will have contributed to this decision. It is important for Qualcomm management to keep labor variances minimal in the future so that large workforce reductions are not required to control costs.

THINK IT THROUGH

Chocolate Cow Ice Cream Company

The Chocolate Cow Ice Cream Company has grown substantially recently, and management now feels the need to develop standards and compute variances. A consulting firm was hired to develop the standards and the format for the variance computation. One standard in particular that the consulting firm developed seemed too excessive to plant management. The consulting firm's standard was production of 100 gallons of ice cream every 45 minutes. The plant's middle level of management thought the standard should be 100 gallons every 55 minutes, while the top management of the company thought that the consulting firm's standard would provide more motivation to the employees.

1. Why is the company establishing a standard for production?
2. What are some factors the company may need to consider before selecting one of the proposed standards?

8.2 Compute and Evaluate Materials Variances

As you've learned, direct materials are those materials used in the production of goods that are easily traceable and are a major component of the product. The amount of materials used and the price paid for those materials may differ from the standard costs determined at the beginning of a period. A company can compute these materials variances and, from these calculations, can interpret the results and decide how to address these differences.

CONCEPTS IN PRACTICE

Buttering Popcorn

In a movie theater, management uses standards to determine if the proper amount of butter is being used on the popcorn. They train the employees to put two tablespoons of butter on each bag of popcorn, so total butter usage is based on the number of bags of popcorn sold. Therefore, if the theater sells 300 bags of popcorn with two tablespoons of butter on each, the total amount of butter that should be used is 600

2 Munsif Vengattil. "Qualcomm Begins Layoffs as Part of Cost Cuts." *Reuters*. April 18, 2018. https://www.reuters.com/article/us-qualcomm-layoffs/qualcomm-begins-layoffs-as-part-of-cost-cuts-idUSKBN1HP33L

tablespoons. Management can then compare the predicted use of 600 tablespoons of butter to the actual amount used. If the actual usage of butter was less than 600, customers may not be happy, because they may feel that they did not get enough butter. If more than 600 tablespoons of butter were used, management would investigate to determine why. Some reasons why more butter was used than expected (unfavorable outcome) would be because of inexperienced workers pouring too much, or the standard was set too low, producing unrealistic expectations that do not satisfy customers.

Fundamentals of Direct Materials Variances

The direct materials variances measure how efficient the company is at using materials as well as how effective it is at using materials. There are two components to a **direct materials variance**, the direct materials price variance and the direct materials quantity variance, which both compare the actual price or amount used to the standard amount.

Direct Materials Price Variance

The **direct materials price variance** compares the actual price per unit (pound or yard, for example) of the direct materials to the standard price per unit of direct materials. The formula for direct materials price variance is calculated as:

$$\text{Direct Materials Price Variance} = \left(\begin{array}{c}\text{Actual Quantity Purchased}\\ \times \\ \text{Actual Price Paid}\end{array}\right) - \left(\begin{array}{c}\text{Actual Quantity Purchased}\\ \times \\ \text{Standard Price}\end{array}\right)$$

Factoring out actual quantity used from both components of the formula, it can be rewritten as:

$$\text{Direct Materials Price Variance} = \left(\begin{array}{c}\text{Actual Price per}\\ \text{Unit of Materials}\end{array} - \begin{array}{c}\text{Standard Price per}\\ \text{Unit of Materials}\end{array}\right) \times \begin{array}{c}\text{Actual Quantity of}\\ \text{Materials Purchased}\end{array}$$

With either of these formulas, the actual quantity purchased refers to the actual amount of materials bought during the period. The standard price is the expected price paid for materials per unit. The actual price paid is the actual amount paid for materials per unit. If there is no difference between the standard price and the actual price paid, the outcome will be zero, and no price variance exists.

If the actual price paid per unit of material is lower than the standard price per unit, the variance will be a favorable variance. A favorable outcome means you spent less on the purchase of materials than you anticipated. If, however, the actual price paid per unit of material is greater than the standard price per unit, the variance will be unfavorable. An unfavorable outcome means you spent more on the purchase of materials than you anticipated.

The actual price can differ from the standard or expected price because of such factors as supply and demand of the material, increased labor costs to the supplier that are passed along to the customer, or improvements in technology that make the material cheaper. The producer must be aware that the difference between what it expects to happen and what actually happens will affect all of the goods produced using these particular materials. Therefore, the sooner management is aware of a problem, the sooner they can fix it. For that reason, the material price variance is computed at the time of purchase and not when the material is used in production.

Let us consider an example. Connie's Candy Company produces various types of candies that they sell to retailers. Connie's Candy establishes a standard price for candy-making materials of $7.00 per pound. Each box of candy is expected to use 0.25 pounds of candy-making materials. Connie's Candy found that the actual price of materials was $6.00 per pound. They purchased 20 pounds of materials and they still actually use 0.25 pounds of materials to make each box. The direct materials price variance computes as:

$$\text{Direct Materials Price Variance} = (\$6.00 - \$7.00) \times 20 \text{ lbs.} = \$20 \text{ or } \$20 \text{ (Favorable)}$$

In this case, the actual price per unit of materials is $6.00, the standard price per unit of materials is $7.00, and the actual quantity purchased is 20 pounds. This computes as a favorable outcome. This is a favorable outcome because the actual price for materials was less than the standard price. As a result of this favorable outcome information, the company may consider continuing operations as they exist, or could change future budget projections to reflect higher profit margins, among other things.

Let us take the same example except now the actual price for candy-making materials is $9.00 per pound. The direct materials price variance computes as:

$$\text{Direct Materials Price Variance} = (\$9.00 - \$7.00) \times 20 \text{ lbs.} = \$40 \text{ or } \$40 \text{ (Unfavorable)}$$

In this case, the actual price per unit of materials is $9.00, the standard price per unit of materials is $7.00, and the actual quantity purchased is 20 pounds. This computes as an unfavorable outcome. This is an unfavorable outcome because the actual price for materials was more than the standard price. As a result of this unfavorable outcome information, the company may consider using cheaper materials, changing suppliers, or increasing prices to cover costs.

Another element this company and others must consider is a direct materials quantity variance.

THINK IT THROUGH

Don't "Skirt" the Issue

You run a fabric store and order materials through a supplier. At the end of the month, you review your materials cost and discover that your direct materials price and quantity variances produced unfavorable results. What could be attributed to these unfavorable outcomes? How would these unfavorable outcomes impact the total direct materials variance?

Direct Materials Quantity Variance

The **direct materials quantity variance** compares the actual quantity of materials used to the standard materials that were expected to be used to make the actual units produced. The variance is calculated using this formula:

$$\text{Direct Materials Quantity Variance} = \left(\begin{array}{c}\text{Actual Quantity Used} \\ \times \\ \text{Standard Price}\end{array}\right) - \left(\begin{array}{c}\text{Standard Quantity} \\ \times \\ \text{Standard Price}\end{array}\right)$$

Factoring out standard price from both components of the formula, it can be rewritten as:

$$\text{Direct Materials Quantity Variance} = \left(\begin{array}{c}\text{Actual Quantity of} \\ \text{Materials Used} \\ \text{for Units Produced}\end{array} - \begin{array}{c}\text{Standard Quantity} \\ \text{of Materials Expected} \\ \text{for the Units Produced}\end{array}\right) \times \begin{array}{c}\text{Standard} \\ \text{Price}\end{array}$$

With either of these formulas, the actual quantity used refers to the actual amount of materials used at the actual production output. The standard price is the expected price paid for materials per unit. The standard quantity is the expected amount of materials used at the actual production output. If there is no difference between the actual quantity used and the standard quantity, the outcome will be zero, and no variance exists.

If the actual quantity of materials used is less than the standard quantity used at the actual production output level, the variance will be a favorable variance. A favorable outcome means you used fewer materials than anticipated, to make the actual number of production units. If, however, the actual quantity of materials used is greater than the standard quantity used at the actual production output level, the variance will be

unfavorable. An unfavorable outcome means you used more materials than anticipated to make the actual number of production units.

The actual quantity used can differ from the standard quantity because of improved efficiencies in production, carelessness or inefficiencies in production, or poor estimation when creating the standard usage.

Consider the previous example with Connie's Candy Company. Connie's Candy established a standard price for candy-making materials of $7.00 per pound. Each box of candy is expected to use 0.25 pounds of candy-making materials. Connie's Candy found that the actual quantity of candy-making materials used to produce one box of candy was 0.20 per pound. The direct materials quantity variance computes as:

$$\text{Direct Materials Quantity Variance} = (0.20\text{ lb.} - 0.25\text{ lb.}) \times \$7.00 = -\$0.35 \text{ or } \$0.35 \text{ (Favorable)}$$

In this case, the actual quantity of materials used is 0.20 pounds, the standard price per unit of materials is $7.00, and the standard quantity used is 0.25 pounds. This computes as a favorable outcome. This is a favorable outcome because the actual quantity of materials used was less than the standard quantity expected at the actual production output level. As a result of this favorable outcome information, the company may consider continuing operations as they exist, or could change future budget projections to reflect higher profit margins, among other things.

Let us take the same example except now the actual quantity of candy-making materials used to produce one box of candy was 0.50 per pound. The direct materials quantity variance computes as:

$$\text{Direct Materials Quantity Variance} = (0.50\text{ lb.} - 0.25\text{ lb.}) \times \$7.00 = \$1.75 \text{ or } \$1.75 \text{ (Unfavorable)}$$

In this case, the actual quantity of materials used is 0.50 pounds, the standard price per unit of materials is $7.00, and the standard quantity used is 0.25 pounds. This computes as an unfavorable outcome. This is an unfavorable outcome because the actual quantity of materials used was more than the standard quantity expected at the actual production output level. As a result of this unfavorable outcome information, the company may consider retraining workers to reduce waste or change their production process to decrease materials needs per box.

The combination of the two variances can produce one overall total direct materials cost variance.

LINK TO LEARNING

Watch this video featuring a professor of accounting walking through the steps involved in calculating a material price variance and a material quantity variance (https://openstax.org/l/50Variances) to learn more.

Total Direct Materials Cost Variance

When a company makes a product and compares the actual materials cost to the standard materials cost, the result is the **total direct materials cost variance**.

$$\text{Total Direct Materials Variance} = \left(\begin{matrix}\text{Actual Quantity} \\ \times \\ \text{Actual Price}\end{matrix}\right) - \left(\begin{matrix}\text{Standard Quantity} \\ \times \\ \text{Standard Price}\end{matrix}\right)$$

An unfavorable outcome means the actual costs related to materials were more than the expected (standard) costs. If the outcome is a favorable outcome, this means the actual costs related to materials are less than the expected (standard) costs.

The total direct materials cost variance is also found by combining the direct materials price variance and the direct materials quantity variance. By showing the total materials variance as the sum of the two components, management can better analyze the two variances and enhance decision-making.

Figure 8.3 shows the connection between the direct materials price variance and direct materials quantity variance to total direct materials cost variance.

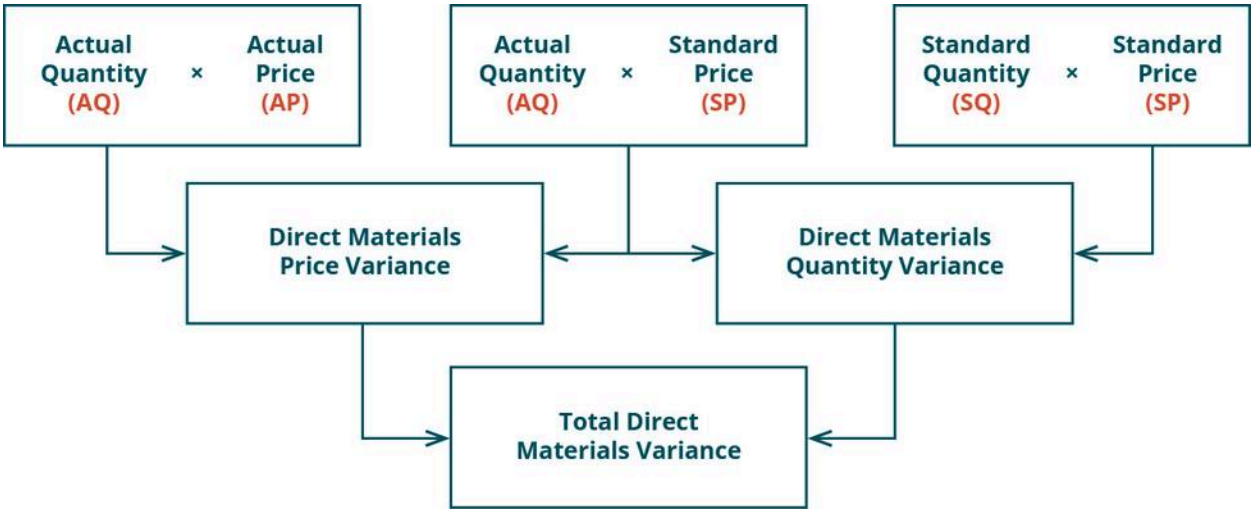


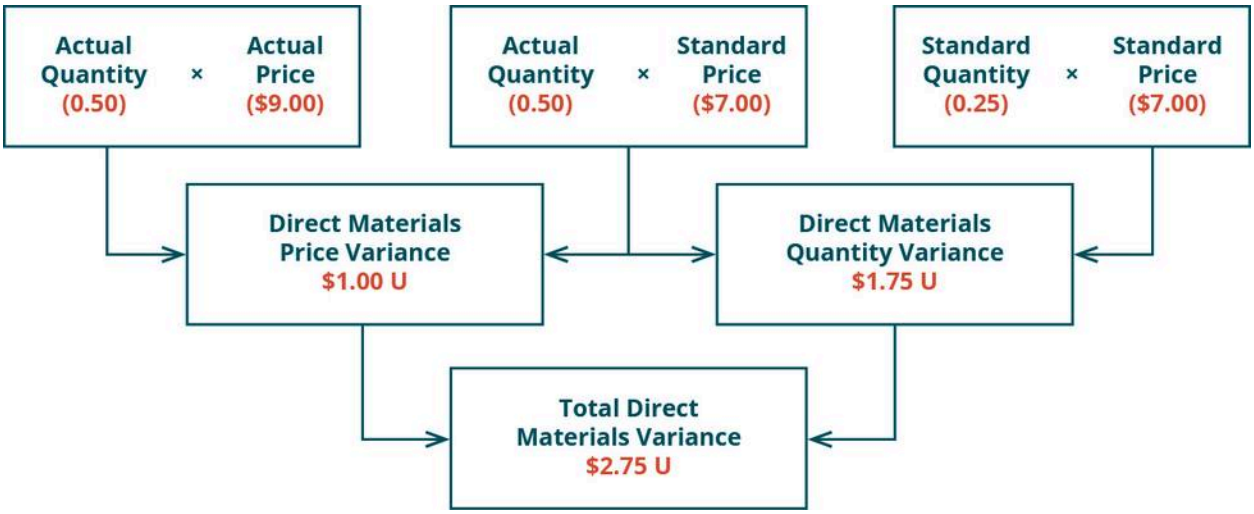
Figure 8.3 Direct Materials Variance. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

For example, Connie's Candy Company expects to pay $7.00 per pound for candy-making materials but actually pays $9.00 per pound. The company expected to use 0.25 pounds of materials per box but actually used 0.50 per box. The total direct materials variance is computed as:

$$\text{Total Direct Materials Variance} = (0.50 \text{ lbs.} \times \$9.00) - (0.25 \text{ lbs.} \times \$7.00) = \$4.50 - \$1.75 = \$2.75 \text{ (Unfavorable)}$$

In this case, two elements contribute to the unfavorable outcome. Connie's Candy paid $2.00 per pound more for materials than expected and used 0.25 pounds more of materials than expected to make one box of candy.

The same calculation is shown using the outcomes of the direct materials price and quantity variances.



As with the interpretations for the materials price and quantity variances, the company would review the individual components contributing to the overall unfavorable outcome for the total direct materials variance, and possibly make changes to production elements as a result.

YOUR TURN

Sweet and Fresh Shampoo Materials

Biglow Company makes a hair shampoo called Sweet and Fresh. Each bottle has a standard material cost of

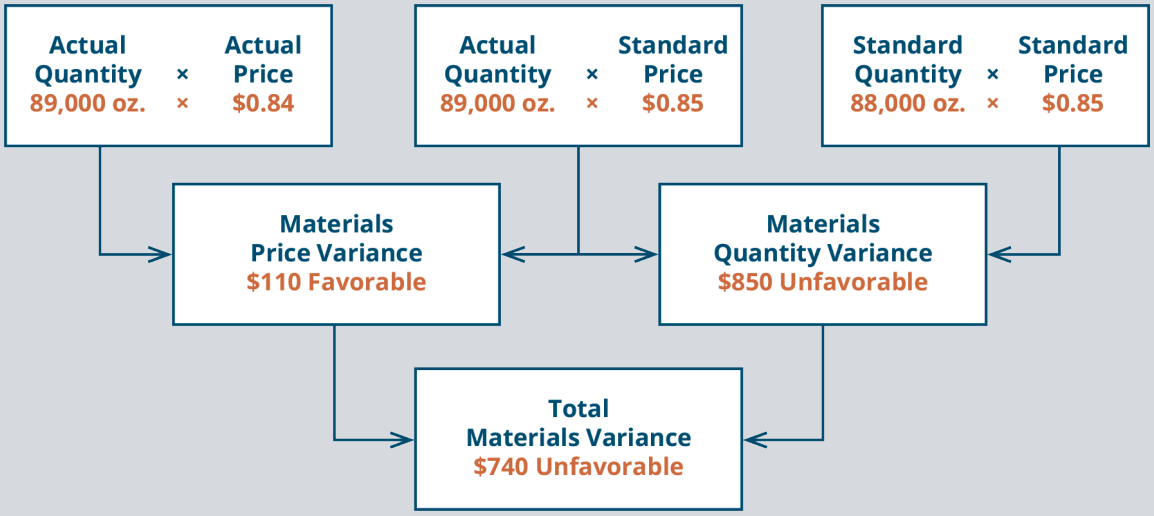
8 ounces at $0.85 per ounce. During May, Biglow manufactured 11,000 bottles. They bought 89,000 ounces of material at a cost of $74,760. All 89,000 ounces were used to make the 11,000 bottles. Calculate the material price variance and the material quantity variance.

Solution

Actual price per pound: 74,760/89,000 = $0.84

Material price variance: 89,000 × (0.84 − 0.85) = $890 favorable

Material quantity variance: 0.85 × (89,000 – 88,000) = $850 unfavorable



8.3 Compute and Evaluate Labor Variances

In addition to evaluating materials usage, companies must assess how efficiently and effectively they are using labor in the production of their products. Direct labor is a cost associated with workers working directly in the production process. The company must look at both the quantity of hours used and the rate of the labor and compare outcomes to standard costs. Determining efficiency and effectiveness of labor leads to individual labor variances. A company can compute these labor variances and make informed decisions about labor operations based on these differences.

Fundamentals of Direct Labor Variances

The **direct labor variance** measures how efficiently the company uses labor as well as how effective it is at pricing labor. There are two components to a labor variance, the direct labor rate variance and the direct labor time variance.

Direct Labor Rate Variance

The **direct labor rate variance** compares the actual rate per hour of direct labor to the standard rate per hour of labor for the hours worked. The direct labor rate variance is calculated using this formula:

$$\text{Direct Labor Rate Variance} = \left(\begin{array}{c}\text{Actual Hours Worked} \\ \times \\ \text{Actual Rate per Hour}\end{array}\right) - \left(\begin{array}{c}\text{Actual Hours Worked} \\ \times \\ \text{Standard Rate per Hour}\end{array}\right)$$

Factoring out the actual hours worked from both components of the formula, it can be rewritten as

$$\text{Direct Labor Rate Variance} = \left(\text{Actual Rate per Hour} - \text{Standard Rate per Hour}\right) \times \text{Actual Hours Worked}$$

With either of these formulas, the actual rate per hour refers to the actual rate of pay for workers to create one unit of product. The standard rate per hour is the expected rate of pay for workers to create one unit of product. The actual hours worked are the actual number of hours worked to create one unit of product. If there is no difference between the standard rate and the actual rate, the outcome will be zero, and no variance exists.

If the actual rate of pay per hour is less than the standard rate of pay per hour, the variance will be a favorable variance. A favorable outcome means you paid workers less than anticipated. If, however, the actual rate of pay per hour is greater than the standard rate of pay per hour, the variance will be unfavorable. An unfavorable outcome means you paid workers more than anticipated.

The actual rate can differ from the standard or expected rate because of supply and demand of the workers, increased labor costs due to economic changes or union contracts, or the ability to hire employees at a different skill level. Once the manufacturer makes the products, the labor costs will follow the goods through production, so the company should evaluate how the difference between what it expected to happen and what actually happened will affect all the goods produced using these particular labor rates.

Let us again consider Connie's Candy Company with respect to labor. Connie's Candy establishes a standard rate per hour for labor of $8.00. Each box of candy is expected to require 0.10 hours of labor (6 minutes). Connie's Candy found that the actual rate of pay per hour for labor was $7.50. They still actually required 0.10 hours of labor to make each box. The direct labor rate variance computes as:

$$\text{Direct Labor Rate Variance} = (\$7.50 - \$8.00) \times 0.10 \text{ hours} = -\$0.05 \text{ or } \$0.05 \text{ (Favorable)}$$

In this case, the actual rate per hour is $7.50, the standard rate per hour is $8.00, and the actual hour worked is 0.10 hours per box. This computes as a favorable outcome. This is a favorable outcome because the actual rate of pay was less than the standard rate of pay. As a result of this favorable outcome information, the company may consider continuing operations as they exist, or could change future budget projections to reflect higher profit margins, among other things.

Let us take the same example except now the actual rate of pay per hour is $9.50. The direct labor rate variance computes as:

$$\text{Direct Labor Rate Variance} = (\$9.50 - \$8.00) \times 0.10 \text{ hours} = \$0.15 \text{ or } \$0.15 \text{ (Unfavorable)}$$

In this case, the actual rate per hour is $9.50, the standard rate per hour is $8.00, and the actual hours worked per box are 0.10 hours. This computes as an unfavorable outcome. This is an unfavorable outcome because the actual rate per hour was more than the standard rate per hour. As a result of this unfavorable outcome information, the company may consider using cheaper labor, changing the production process to be more efficient, or increasing prices to cover labor costs.

Another element this company and others must consider is a direct labor time variance.

Direct Labor Time Variance

The **direct labor time variance** compares the actual labor hours used to the standard labor hours that were expected to be used to make the actual units produced. The variance is calculated using this formula:

$$\text{Direct Labor Time Variance} = \left(\begin{matrix}\text{Actual Hours Worked} \\ \times \\ \text{Standard Rate per Hour}\end{matrix}\right) - \left(\begin{matrix}\text{Standard Hours} \\ \times \\ \text{Standard Rate per Hour}\end{matrix}\right)$$

Factoring out the standard rate per hour from both components of the formula, it can be rewritten as:

$$\text{Direct Labor Time Variance} = \left(\text{Actual Hours Worked} - \text{Standard Hours Expected for the Units Produced}\right) \times \text{Standard Rate per Hour}$$

With either of these formulas, the actual hours worked refers to the actual number of hours used at the actual production output. The standard rate per hour is the expected hourly rate paid to workers. The standard hours are the expected number of hours used at the actual production output. If there is no difference between the actual hours worked and the standard hours, the outcome will be zero, and no variance exists.

If the actual hours worked are less than the standard hours at the actual production output level, the variance will be a favorable variance. A favorable outcome means you used fewer hours than anticipated to make the actual number of production units. If, however, the actual hours worked are greater than the standard hours at the actual production output level, the variance will be unfavorable. An unfavorable outcome means you used more hours than anticipated to make the actual number of production units.

The actual hours used can differ from the standard hours because of improved efficiencies in production, carelessness or inefficiencies in production, or poor estimation when creating the standard usage.

Consider the previous example with Connie's Candy Company. Connie's Candy establishes a standard rate per hour for labor of $8.00. Each box of candy is expected to require 0.10 hours of labor (6 minutes). Connie's Candy found that the actual hours worked per box were 0.05 hours (3 minutes). The actual rate per hour for labor remained at $8.00 to make each box. The direct labor time variance computes as:

$$\text{Direct Labor Time Variance} = (0.05 - 0.10) \times \$8.00 \text{ per hour} = -\$0.40 \text{ or } \$0.40 \text{ (Favorable)}$$

In this case, the actual hours worked are 0.05 per box, the standard hours are 0.10 per box, and the standard rate per hour is $8.00. This computes as a favorable outcome. This is a favorable outcome because the actual hours worked were less than the standard hours expected. As a result of this favorable outcome information, the company may consider continuing operations as they exist, or could change future budget projections to reflect higher profit margins, among other things.

Let us take the same example except now the actual hours worked are 0.20 hours per box. The direct labor time variance computes as:

$$\text{Direct Labor Time Variance} = (\$0.20 - \$0.10) \times \$8.00 \text{ per hour} = \$0.80 \text{ or } \$0.80 \text{ (Unfavorable)}$$

In this case, the actual hours worked per box are 0.20, the standard hours per box are 0.10, and the standard rate per hour is $8.00. This computes as an unfavorable outcome. This is an unfavorable outcome because the actual hours worked were more than the standard hours expected per box. As a result of this unfavorable outcome information, the company may consider retraining its workers, changing the production process to be more efficient, or increasing prices to cover labor costs.

The combination of the two variances can produce one overall total direct labor cost variance.

THINK IT THROUGH

Package Deliveries

UPS drivers are evaluated on how many miles they drive and how quickly they deliver packages. The drivers are given the route and time they are expected to take, so they are expected to complete their route in a timely and efficient manner. They also work until all packages are delivered. A GPS tracking system tracks the trucks throughout the day. The system keeps track of how much they back up and if they take any left turns because right turns are much more time efficient.[3] Tracking drivers like this does not leave them very much time to deal with customers. Customer service is a major part of the driver's job. Can the driver service the customer and drive the route in the time and distance allotted? Which is more important: customer service or driving the route in a timely and efficient manner?

LINK TO LEARNING

Watch this video presenting an instructor walking through the steps involved in calculating direct labor variances (https://openstax.org/l/50DirLaborVar) to learn more.

Total Direct Labor Variance

When a company makes a product and compares the actual labor cost to the standard labor cost, the result is the **total direct labor variance**.

$$\text{Total Direct Labor Variance} = (\text{Actual Hours} \times \text{Actual Rate}) - (\text{Standard Hours} \times \text{Standard Rate})$$

If the outcome is unfavorable, the actual costs related to labor were more than the expected (standard) costs. If the outcome is favorable, the actual costs related to labor are less than the expected (standard) costs.

The total direct labor variance is also found by combining the direct labor rate variance and the direct labor time variance. By showing the total direct labor variance as the sum of the two components, management can better analyze the two variances and enhance decision-making.

Figure 8.4 shows the connection between the direct labor rate variance and direct labor time variance to total direct labor variance.

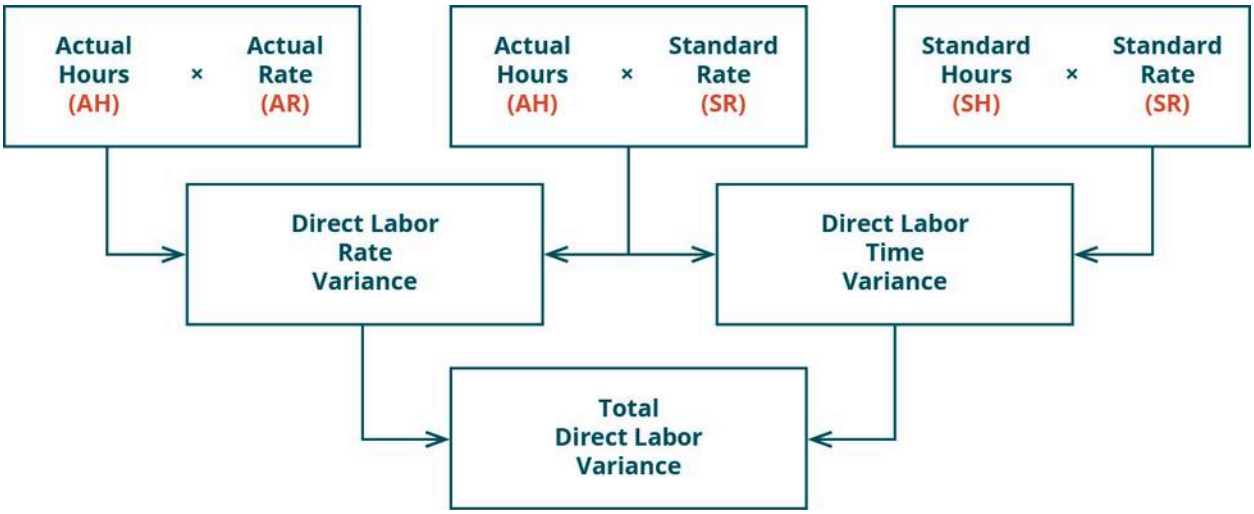


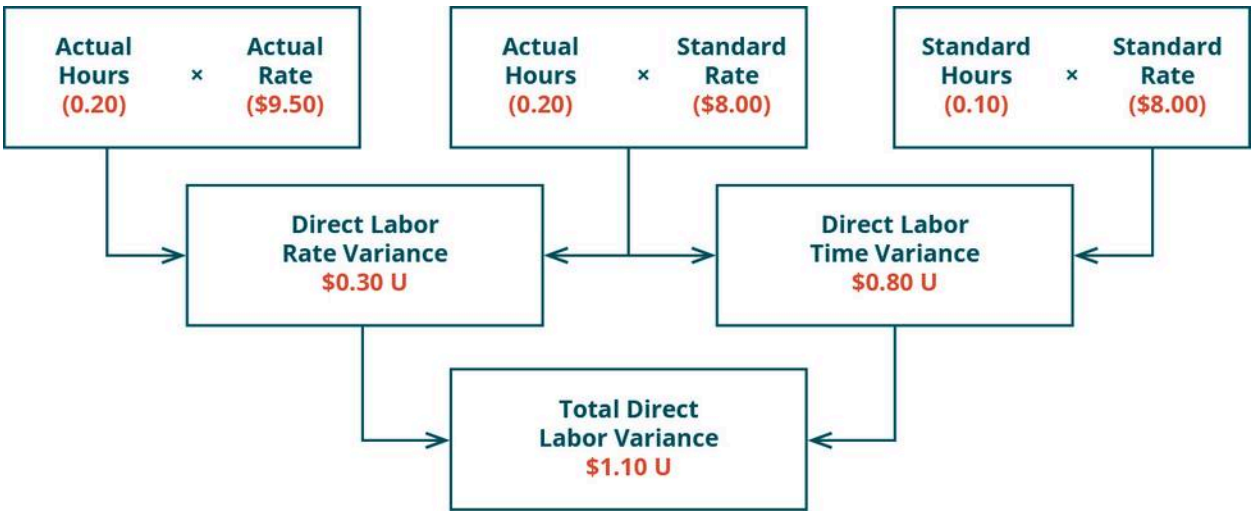
Figure 8.4 Direct Labor Variance. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

For example, Connie's Candy Company expects to pay a rate of $8.00 per hour for labor but actually pays $9.50 per hour. The company expected to use 0.10 hours of labor per box but actually used 0.20 hours per box. The total direct labor variance is computed as:

$$\text{Total Direct Labor Time Variance} = (0.20 \text{ hours} \times \$9.50) - (0.10 \text{ hours} \times \$8.00) = \$1.90 - \$0.80 = \$1.10 \text{ (Unfavorable)}$$

In this case, two elements are contributing to the unfavorable outcome. Connie's Candy paid $1.50 per hour more for labor than expected and used 0.10 hours more than expected to make one box of candy. The same calculation is shown as follows using the outcomes of the direct labor rate and time variances.

3 Graham Kendall. "Why UPS Drivers Don't Turn Left and Why You Shouldn't Either." *The Conversation*. January 20, 2017. http://theconversation.com/why-ups-drivers-dont-turn-left-and-you-probably-shouldnt-either-71432



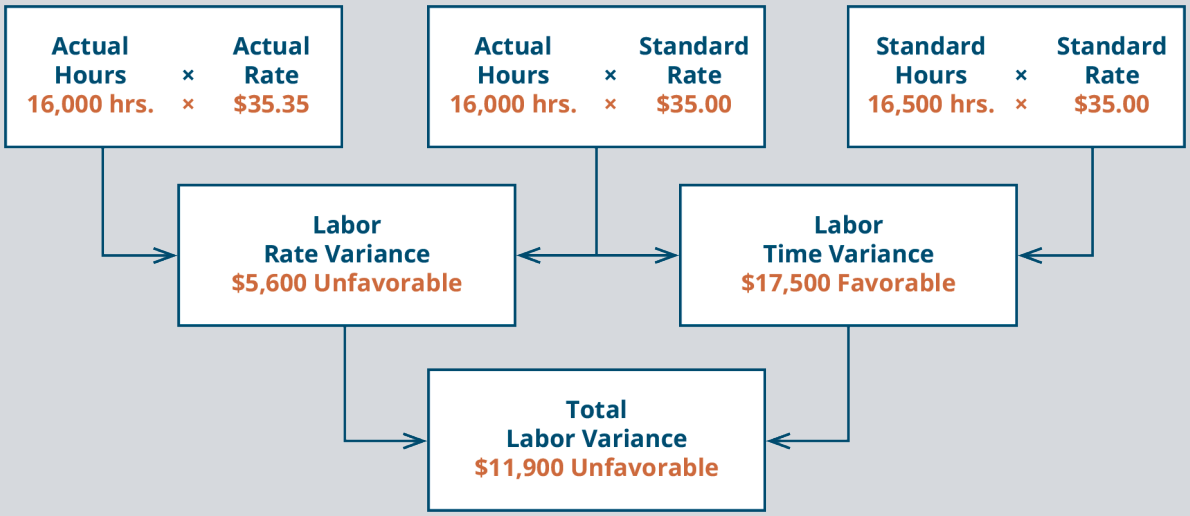
As with the interpretations for the labor rate and time variances, the company would review the individual components contributing to the overall unfavorable outcome for the total direct labor variance, and possibly make changes to production elements as a result.

YOUR TURN

Sweet and Fresh Shampoo Labor

Biglow Company makes a hair shampoo called Sweet and Fresh. Each bottle has a standard labor cost of 1.5 hours at $35.00 per hour. During May, Biglow manufactured 11,000 bottles. They used 16,000 hours at a cost of $565,600. Calculate the labor rate variance, labor time variance, and total labor variance.

Solution



CONCEPTS IN PRACTICE

Labor Costs in Service Industries

In the service industry, labor is the main cost. Doctors, for example, have a time allotment for a physical exam and base their fee on the expected time. Insurance companies pay doctors according to a set schedule, so they set the labor standard. They pay a set rate for a physical exam, no matter how long it

takes. If the exam takes longer than expected, the doctor is not compensated for that extra time. This would produce an unfavorable labor variance for the doctor. Doctors know the standard and try to schedule accordingly so a variance does not exist. If anything, they try to produce a favorable variance by seeing more patients in a quicker time frame to maximize their compensation potential.

8.4 Compute and Evaluate Overhead Variances

Recall that the standard cost of a product includes not only materials and labor but also variable and fixed overhead. It is likely that the amounts determined for standard overhead costs will differ from what actually occurs. This will lead to overhead variances.

Determination and Evaluation of Overhead Variance

In a standard cost system, overhead is applied to the goods based on a standard overhead rate. This is similar to the predetermined overhead rate used previously. The standard overhead rate is calculated by dividing budgeted overhead at a given level of production (known as normal capacity) by the level of activity required for that particular level of production.

$$\text{Standard Overhead Rate} = \frac{\text{Budgeted Overhead Rate}}{\text{Level of Activity}}$$

Usually, the level of activity is either direct labor hours or direct labor cost, but it could be machine hours or units of production.

Creation of Flexible Overhead Budget

To determine the overhead standard cost, companies prepare a **flexible budget** that gives estimated revenues and costs at varying levels of production. The standard overhead cost is usually expressed as the sum of its component parts, fixed and variable costs per unit. Note that at different levels of production, total fixed costs are the same, so the standard fixed cost per unit will change for each production level. However, the variable standard cost per unit is the same per unit for each level of production, but the total variable costs will change.

We continue to use Connie's Candy Company to illustrate. Suppose Connie's Candy budgets capacity of production at 100% and determines expected overhead at this capacity. Connie's Candy also wants to understand what overhead cost outcomes will be at 90% capacity and 110% capacity. The following information is the flexible budget Connie's Candy prepared to show expected overhead at each capacity level.

Percent of capacity	90%	100%	110%
Direct labor hours	1,800	2,000	2,200
Units of output	900	1,000	1,100
Variable overhead	$3,600	$ 4,000	$ 4,400
Fixed overhead	$6,000	$ 6,000	$ 6,000
Total overhead	$9,600	$10,000	$10,400

Normal capacity = 100% and overhead is applied based on direct labor hours
Standard Overhead Rate = $10,000/2,000 = $5 per direct labor hour

Units of output at 100% is 1,000 candy boxes (units). The standard overhead rate is the total budgeted overhead of $10,000 divided by the level of activity (direct labor hours) of 2,000 hours. Notice that fixed overhead remains constant at each of the production levels, but variable overhead changes based on unit output. If Connie's Candy only produced at 90% capacity, for example, they should expect total overhead to be $9,600 and a standard overhead rate of $5.33 (rounded). If Connie's Candy produced 2,200 units, they should

expect total overhead to be $10,400 and a standard overhead rate of $4.73 (rounded). In addition to the total standard overhead rate, Connie's Candy will want to know the variable overhead rates at each activity level.

Using the flexible budget, we can determine the standard variable cost per unit at each level of production by taking the total expected variable overhead divided by the level of activity, which can still be direct labor hours or machine hours.

$$\text{Variable Overhead Rate} = \frac{\text{Budgeted Variable Overhead}}{\text{Level of Activity}}$$

Looking at Connie's Candies, the following table shows the variable overhead rate at each of the production capacity levels.

Production Capacity	Variable/Unit
90%	$3,600/1,800 = $2
100%	$4,000/2,000 = $2
110%	$4,400/2,200 = $2

Sometimes these flexible budget figures and overhead rates differ from the actual results, which produces a variance.

Determination of Variable Overhead Variances

There are two components to variable overhead rates: the overhead application rate and the activity level against which that rate was applied. If we compare the actual variable overhead to the standard variable overhead, by analyzing the difference between actual overhead costs and the standard overhead for current production, it is difficult to determine if the variance is due to application rate differences or activity level differences. Thus, there are two variable overhead variances that will better provide these answers: the variable overhead rate variance and the variable overhead efficiency variance.

Determination of Variable Overhead Rate Variance

The **variable overhead rate variance**, also known as the spending variance, is the difference between the actual variable manufacturing overhead and the variable overhead that was expected given the number of hours worked. The variable overhead rate variance is calculated using this formula:

$$\text{Variable Overhead Rate Variance} = \left(\begin{array}{c}\text{Actual Hours Worked}\\ \times \\ \text{Actual Variable Overhead Rate per Hour}\end{array}\right) - \left(\begin{array}{c}\text{Actual Hours Worked}\\ \times \\ \text{Standard Variable Overhead Rate per Hour}\end{array}\right)$$

Factoring out actual hours worked, we can rewrite the formula as

$$\text{Variable Overhead Rate Variance} = \left(\text{Actual Variable Overhead Rate} - \text{Standard Variable Overhead Rate}\right) \times \text{Actual Hours Worked}$$

If the outcome is favorable (a negative outcome occurs in the calculation), this means the company spent less than what it had anticipated for variable overhead. If the outcome is unfavorable (a positive outcome occurs in the calculation), this means the company spent more than what it had anticipated for variable overhead.

Connie's Candy Company wants to determine if its variable overhead spending was more or less than anticipated. Connie's Candy had this data available in the flexible budget:

Percent of capacity	100%
Direct labor hours	2,000
Units of output	1,000
Variable overhead	$ 4,000
Fixed overhead	$ 6,000
Total overhead	$10,000

Connie's Candy also had this actual output information:

Percent of capacity	100%
Direct labor hours	2,500
Units of output	1,000
Variable overhead	$ 7,000
Fixed overhead	$ 6,000
Total overhead	$13,000

To determine the variable overhead rate variance, the standard variable overhead rate per hour and the actual variable overhead rate per hour must be determined. The standard variable overhead rate per hour is $2.00 ($4,000/2,000 hours), taken from the flexible budget at 100% capacity. The actual variable overhead rate is $2.80 ($7,000/2,500), taken from the actual results at 100% capacity. Therefore,

$$\text{Variable Overhead Rate Variance} = (\$2.80 - \$2.00) \times 2{,}500 = \$2{,}000 \text{ (Unfavorable)}$$

This produces an unfavorable outcome. This could be for many reasons, and the production supervisor would need to determine where the variable cost difference is occurring to make production changes.

Let us look at another example producing a favorable outcome. Connie's Candy had this data available in the flexible budget:

Percent of capacity	100%
Direct labor hours	2,000
Units of output	1,000
Variable overhead	$ 4,000
Fixed overhead	$ 6,000
Total overhead	$10,000

Connie's Candy also had this actual output information:

Percent of capacity	100%
Direct labor hours	2,000
Units of output	1,000
Variable overhead	$3,500
Fixed overhead	$6,000
Total overhead	$9,500

To determine the variable overhead rate variance, the standard variable overhead rate per hour and the actual variable overhead rate per hour must be determined. The standard variable overhead rate per hour is $2.00 ($4,000/2,000 hours), taken from the flexible budget at 100% capacity. The actual variable overhead rate is $1.75 ($3,500/2,000), taken from the actual results at 100% capacity. Therefore,

$$\text{Variable Overhead Rate Variance} = (\$1.75 - \$2.00) \times \$2{,}000 = -\$500 \text{ or } \$500 \text{ (Favorable)}$$

This produces a favorable outcome. This could be for many reasons, and the production supervisor would need to determine where the variable cost difference is occurring to better understand the variable overhead reduction.

Interpretation of the variable overhead rate variance is often difficult because the cost of one overhead item,

such as indirect labor, could go up, but another overhead cost, such as indirect materials, could go down. Often, explanation of this variance will need clarification from the production supervisor. Another variable overhead variance to consider is the variable overhead efficiency variance.

Determination of Variable Overhead Efficiency Variance

The **variable overhead efficiency variance**, also known as the controllable variance, is driven by the difference between the actual hours worked and the standard hours expected for the units produced. This variance measures whether the allocation base was efficiently used. The variable overhead efficiency variance is calculated using this formula:

$$\text{Variable Overhead Efficiency Variance} = \left(\begin{array}{c}\text{Actual Hours Worked}\\ \times \\ \text{Standard Variable Overhead Rate per Hour}\end{array}\right) - \left(\begin{array}{c}\text{Standard Hours}\\ \times \\ \text{Standard Variable Overhead Rate per Hour}\end{array}\right)$$

Factoring out standard overhead rate, the formula can be written as

$$\text{Variable Overhead Efficiency Variance} = (\text{Actual Labor Hours} - \text{Standard Labor Hours}) \times \text{Standard Overhead Rate}$$

If the outcome is favorable (a negative outcome occurs in the calculation), this means the company was more efficient than what it had anticipated for variable overhead. If the outcome is unfavorable (a positive outcome occurs in the calculation), this means the company was less efficient than what it had anticipated for variable overhead.

Connie's Candy Company wants to determine if its variable overhead efficiency was more or less than anticipated. Connie's Candy had the following data available in the flexible budget:

Percent of capacity	**100%**
Direct labor hours	2,000
Units of output	1,000
Variable overhead	$ 4,000
Fixed overhead	$ 6,000
Total overhead	$10,000

Connie's Candy also had the following actual output information:

Percent of capacity	**100%**
Direct labor hours	2,500
Units of output	1,000
Variable overhead	$ 7,000
Fixed overhead	$ 6,000
Total overhead	$13,000

To determine the variable overhead efficiency variance, the actual hours worked and the standard hours worked at the production capacity of 100% must be determined. Actual hours worked are 2,500, and standard hours are 2,000. The standard variable overhead rate per hour is $2.00 ($4,000/2,000 hours), taken from the flexible budget at 100% capacity. Therefore,

$$\text{Variable Overhead Efficiency Variance} = (2{,}500 - 2{,}000) \times \$2.00 = \$1{,}000 \text{ (Unfavorable)}$$

This produces an unfavorable outcome. This could be for many reasons, and the production supervisor would need to determine where the variable cost difference is occurring to make production changes.

Let us look at another example producing a favorable outcome. Connie's Candy had the following data available in the flexible budget:

Percent of capacity	100%
Direct labor hours	2,000
Units of output	1,000
Variable overhead	$ 4,000
Fixed overhead	$ 6,000
Total overhead	$10,000

Connie's Candy also had the following actual output information:

Percent of capacity	100%
Direct labor hours	1,800
Units of output	1,000
Variable overhead	$3,500
Fixed overhead	$6,000
Total overhead	$9,500

To determine the variable overhead efficiency variance, the actual hours worked and the standard hours worked at the production capacity of 100% must be determined. Actual hours worked are 1,800, and standard hours are 2,000. The standard variable overhead rate per hour is $2.00 ($4,000/2,000 hours), taken from the flexible budget at 100% capacity. Therefore,

$$\text{Variable Overhead Efficiency Variance} = (1{,}800 - 2{,}000) \times \$2.00 = -\$400 \text{ or } \$400 \text{ (Favorable)}$$

This produces a favorable outcome. This could be for many reasons, and the production supervisor would need to determine where the variable cost difference is occurring to better understand the variable overhead efficiency reduction.

The **total variable overhead cost variance** is also found by combining the variable overhead rate variance and the variable overhead efficiency variance. By showing the total variable overhead cost variance as the sum of the two components, management can better analyze the two variances and enhance decision-making.

Figure 8.5 shows the connection between the variable overhead rate variance and variable overhead efficiency variance to total variable overhead cost variance.

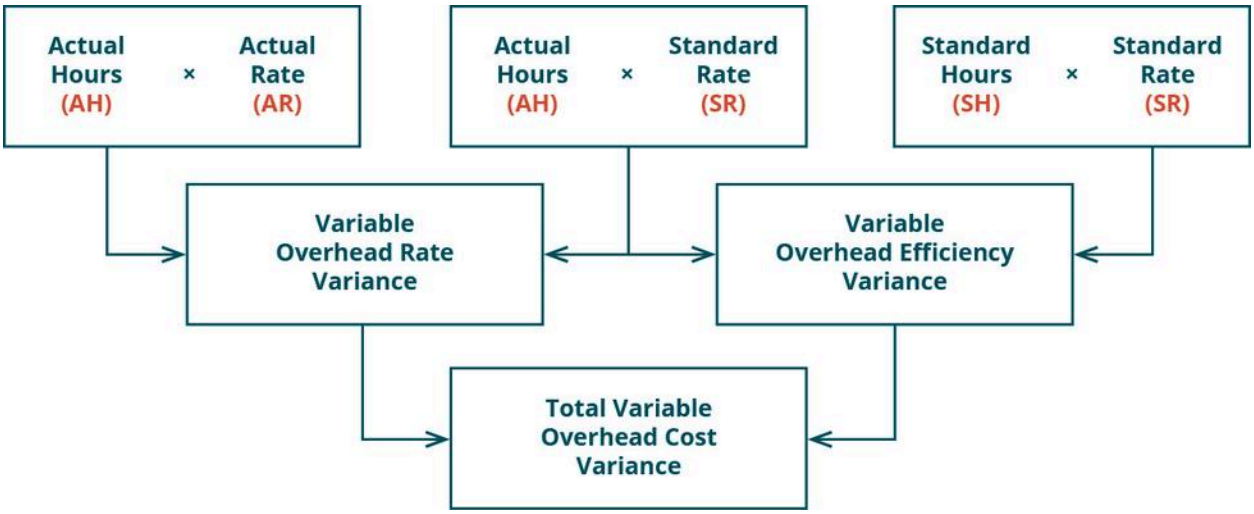


Figure 8.5 Variable Overheard Cost Variance. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

For example, Connie's Candy Company had the following data available in the flexible budget:

Percent of capacity	100%
Direct labor hours	2,000
Units of output	1,000
Variable overhead	$ 4,000
Fixed overhead	$ 6,000
Total overhead	$10,000

Connie's Candy also had the following actual output information:

Percent of capacity	100%
Direct labor hours	1,800
Units of output	1,000
Variable overhead	$3,500
Fixed overhead	$6,000
Total overhead	$9,500

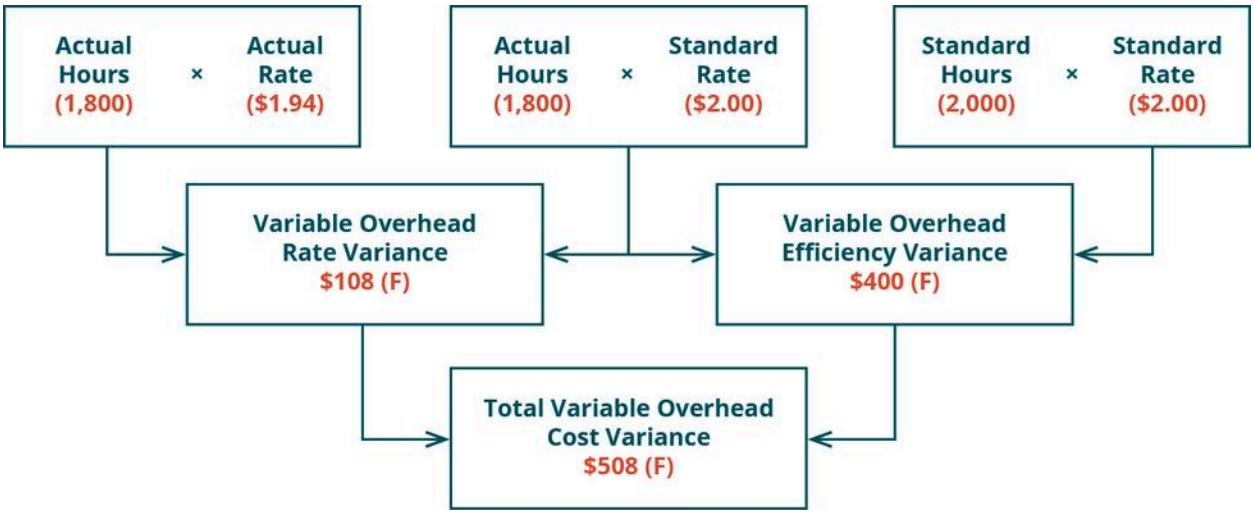
The variable overhead rate variance is calculated as (1,800 × $1.94) – (1,800 × $2.00) = –$108, or $108 (favorable). The variable overhead efficiency variance is calculated as (1,800 × $2.00) – (2,000 × $2.00) = –$400, or $400 (favorable).

The total variable overhead cost variance is computed as:

$$\text{Total Variable Overhead Cost Variance} = (-\$108) + (-\$400) = -\$508 \text{ or } \$508 \text{ (Favorable)}$$

In this case, two elements are contributing to the favorable outcome. Connie's Candy used fewer direct labor hours and less variable overhead to produce 1,000 candy boxes (units).

The same calculation is shown as follows in diagram format.



As with the interpretations for the variable overhead rate and efficiency variances, the company would review the individual components contributing to the overall favorable outcome for the total variable overhead cost variance, before making any decisions about production in the future. Other variances companies consider are fixed factory overhead variances.

Fundamentals of Fixed Factory Overhead Variances

The **fixed factory overhead variance** represents the difference between the actual fixed overhead and the applied fixed overhead. There are two fixed overhead variances. One variance determines if too much or too little was spent on fixed overhead. The other variance computes whether or not actual production was above or below the expected production level.

YOUR TURN

Sweet and Fresh Shampoo Overhead

Biglow Company makes a hair shampoo called Sweet and Fresh. They have the following flexible budget data:

	90%	100%	110%
Direct labor hours	14,000	16,000	18,000
Units of output	10,000	10,000	10,000
Direct labor	$525,000	$ 346,500	$ 378,000
Variable overhead	$315,000	$ 346,000	$ 378,000
Fixed overhead	$ 45,500	$ 45,500	$ 45,500
Total	$953,500	$1,044,300	$1,135,100

What is the standard variable overhead rate at 90%, 100%, and 110% capacity levels?

Solution

90% = $315,000/14,000 = $22.50, 100% = $346,000/16,000 = $21.63 (rounded), 110% = $378,000/18,000 = $21.00.

THINK IT THROUGH

Purchasing Planes

The XYZ Firm is bidding on a contract for a new plane for the military. As the management team is going over the bid, they come to the conclusion it is too high on a per-plane basis, but they cannot find any costs they feel can be reduced. The information from the military states they will purchase between 50 and 100 planes, but will more likely purchase 50 planes rather than 100 planes. XYZ's bid is based on 50 planes. The controller suggests that they base their bid on 100 planes. This would spread the fixed costs over more planes and reduce the bid price. The lower bid price will increase substantially the chances of XYZ winning the bid. Should XYZ Firm keep the bid at 50 planes or increase its bid to 100 planes? What are the pros and cons to keeping the bid at 50 or increasing to 100 planes?

8.5 Describe How Companies Use Variance Analysis

Companies use variance analysis in different ways. The starting point is the determination of standards against which to compare actual results. Many companies produce variance reports, and the management responsible for the variances must explain any variances outside of a certain range. Some companies only require that unfavorable variances be explained, while many companies require both favorable and unfavorable variances to be explained.

Requiring managers to determine what caused unfavorable variances forces them to identify potential problem areas or consider if the variance was a one-time occurrence. Requiring managers to explain favorable variances allows them to assess whether the favorable variance is sustainable. Knowing what caused the favorable variance allows management to plan for it in the future, depending on whether it was a one-time variance or it will be ongoing.

Another possibility is that management may have built the favorable variance into the standards. Management may overestimate the material price, labor rate, material quantity, or labor hours per unit, for

example. This method of overestimation, sometimes called *budget slack*, is built into the standards so management can still look good even if costs are higher than planned. In either case, managers potentially can help other managers and the company overall by noticing particular problem areas or by sharing knowledge that can improve variances.

Often, management will manage "to the variances," meaning they will make decisions that may not be advantageous to the company's best interests over the long run, in order to meet the variance report threshold limits. This can occur when the standards are improperly established, causing significant differences between actual and standard numbers.

ETHICAL CONSIDERATIONS

Ethical Long-Term Decisions in Variance Analysis

The proper use of variance analysis is a significant tool for an organization to reach its long-term goals. When its accounting system recognizes a variance, an organization needs to understand the significant influence of accounting not only in recording its financial results, but also in how reacting to that variance can shape management's behavior toward reaching its goals.[4] Many managers use variance analysis only to determine a short-term reaction, and do not analyze why the variance occurred from a long-term perspective. A more long-term analysis of variances allows an approach that "is responsibility accounting in which authority and accountability for tasks is delegated downward to those managers with the most influence and control over them."[5] It is important for managers to analyze the reported variances with more than just a short-term perspective.

Managers sometimes focus only on making numbers for the current period. For example, a manager might decide to make a manufacturing division's results look profitable in the short term at the expense of reaching the organization's long-term goals. A recognizable cost variance could be an increase in repair costs as a percentage of sales on an increasing basis. This variance could indicate that equipment is not operating efficiently and is increasing overall cost. However, the expense of implementing new, more efficient equipment might be higher than repairing the current equipment. In the short term, it might be more economical to repair the outdated equipment, but in the long term, purchasing more efficient equipment would help the organization reach its goal of eco-friendly manufacturing. If the system use for controlling costs is not aligned to reinforce management of the organization with a long-term perspective, "the manager has no organizational incentive to be concerned with important issues unrelated to anything but the immediate costs"[6] related to the variance. A manager needs to be cognizant of his or her organization's goals when making decisions based on variance analysis.

Management can use standard costs to prepare the budget for the upcoming period, using the past information to possibly make changes to production elements. Standard costs are a measurement tool and can thus be used to evaluate performance. As you've learned, management may manage "to the variances" and can manipulate results to meet expectations. To reduce this possibility, performance should be measured on multiple outcomes, not simply on standard cost variances.

As shown in Table 8.1, standard costs have pros and cons to consider when using them in the decision-making and evaluation processes.

4 Jeffrey R. Cohen and Laurie W. Pant. "The Only Thing That Counts Is That Which Is Counted: A Discussion of Behavioral and Ethical Issues in Cost Accounting That Are Relevant for the OB Professor." September 18, 2018. http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.1026.5569&rep=rep1&type=pdf
5 Jeffrey R. Cohen and Laurie W. Pant. "The Only Thing That Counts Is That Which Is Counted: A Discussion of Behavioral and Ethical Issues in Cost Accounting That Are Relevant for the OB Professor." September 18, 2018. http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.1026.5569&rep=rep1&type=pdf
6 Jeffrey R. Cohen and Laurie W. Pant. "The Only Thing That Counts Is That Which Is Counted: A Discussion of Behavioral and Ethical Issues in Cost Accounting That Are Relevant for the OB Professor." September 18, 2018. http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.1026.5569&rep=rep1&type=pdf

Standard Costs

Pros	Cons
• Useful when developing a future budget • Can be used as a benchmark for performance and quality expectations • Can individually identify areas of success and areas for improvement	• Might ignore customer and employee satisfaction rates • Information could be historical data and not useful in real-time decision-making needs • The system to manage and develop standard costs requires a lot of resources, which could be costly and time consuming

Table 8.1

Standard costing provides many benefits and challenges, and a thorough analysis of each variance and the possible unfavorable or favorable outcomes is required to set future expectations and adjust current production goals.

The following is a summary of all direct materials variances (Figure 8.6), direct labor variances (Figure 8.7), and overhead variances (Figure 8.8) presented as both formulas and tree diagrams. Note that for some of the formulas, there are two presentations of the same formula, for example, there are two presentations of the direct materials price variance. While both arrive at the same answer, students usually prefer one formula structure over the other.

$$\text{Direct Materials Price Variance} = \left(\begin{matrix}\text{Actual Quantity Used} \\ \times \\ \text{Actual Price Paid}\end{matrix}\right) - \left(\begin{matrix}\text{Actual Quantity Used} \\ \times \\ \text{Standard Price}\end{matrix}\right)$$

$$\text{Direct Materials Price Variance} = \left(\begin{matrix}\text{Actual Price per} \\ \text{Unit of Materials}\end{matrix} - \begin{matrix}\text{Standard Price per} \\ \text{Unit of Materials}\end{matrix}\right) \times \begin{matrix}\text{Actual Quantity of} \\ \text{Materials Used}\end{matrix}$$

$$\text{Direct Materials Quantity Variance} = \left(\begin{matrix}\text{Actual Quantity Used} \\ \times \\ \text{Standard Price}\end{matrix}\right) - \left(\begin{matrix}\text{Standard Quantity} \\ \times \\ \text{Standard Price}\end{matrix}\right)$$

$$\text{Direct Materials Quantity Variance} = \left(\begin{matrix}\text{Actual Quantity of} \\ \text{Materials Used} \\ \text{for Units Produced}\end{matrix} - \begin{matrix}\text{Standard Quantity} \\ \text{of Materials Expected} \\ \text{for the Units Produced}\end{matrix}\right) \times \begin{matrix}\text{Standard} \\ \text{Price}\end{matrix}$$

$$\text{Total Direct Materials Variance} = \left(\begin{matrix}\text{Actual Quantity} \\ \times \\ \text{Actual Price}\end{matrix}\right) - \left(\begin{matrix}\text{Standard Quantity} \\ \times \\ \text{Standard Price}\end{matrix}\right)$$

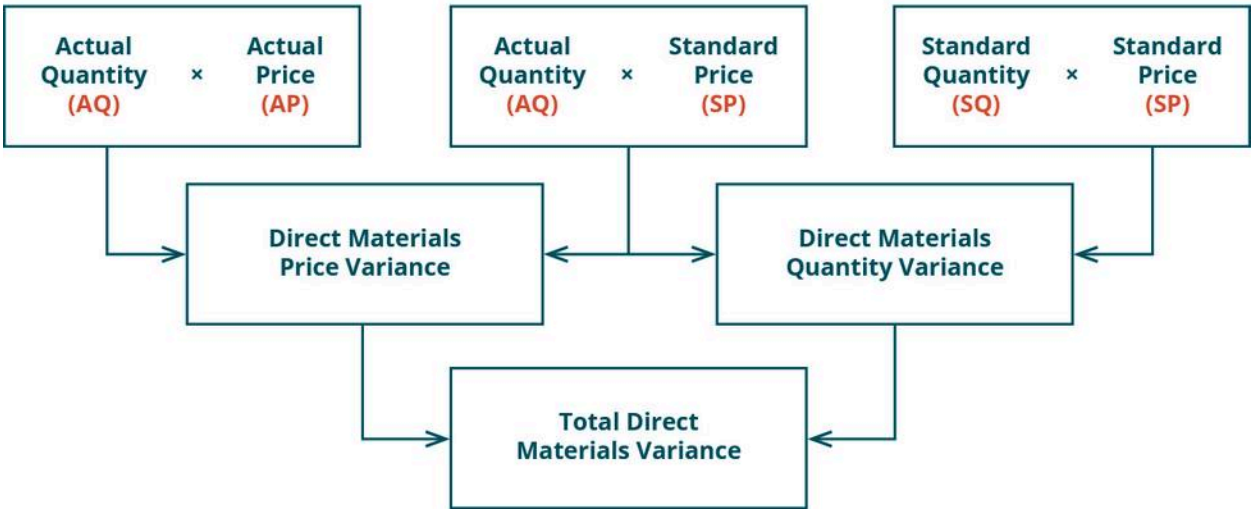


Figure 8.6 Direct Materials Variances. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

$$\text{Direct Labor Rate Variance} = \left(\begin{matrix}\text{Actual Hours Worked} \\ \times \\ \text{Actual Rate per Hour}\end{matrix}\right) - \left(\begin{matrix}\text{Actual Hours Worked} \\ \times \\ \text{Standard Rate per Hour}\end{matrix}\right)$$

$$\text{Direct Labor Rate Variance} = \left(\begin{matrix}\text{Actual Rate} \\ \text{per Hour}\end{matrix} - \begin{matrix}\text{Standard Rate} \\ \text{per Hour}\end{matrix}\right) \times \begin{matrix}\text{Actual Hours} \\ \text{Worked}\end{matrix}$$

$$\text{Direct Labor Time Variance} = \left(\begin{matrix}\text{Actual Hours Worked} \\ \times \\ \text{Standard Rate per Hour}\end{matrix}\right) - \left(\begin{matrix}\text{Standard Hours} \\ \times \\ \text{Standard Rate per Hour}\end{matrix}\right)$$

$$\text{Direct Labor Time Variance} = \left(\begin{matrix}\text{Actual Hours} \\ \text{Worked}\end{matrix} - \begin{matrix}\text{Standard Hours Expected} \\ \text{for the Units Produced}\end{matrix}\right) \times \begin{matrix}\text{Standard Rate} \\ \text{per Hour}\end{matrix}$$

$$\text{Total Direct Labor Variance} = (\text{Actual Hours} \times \text{Actual Rate}) - (\text{Standard Hours} \times \text{Standard Rate})$$

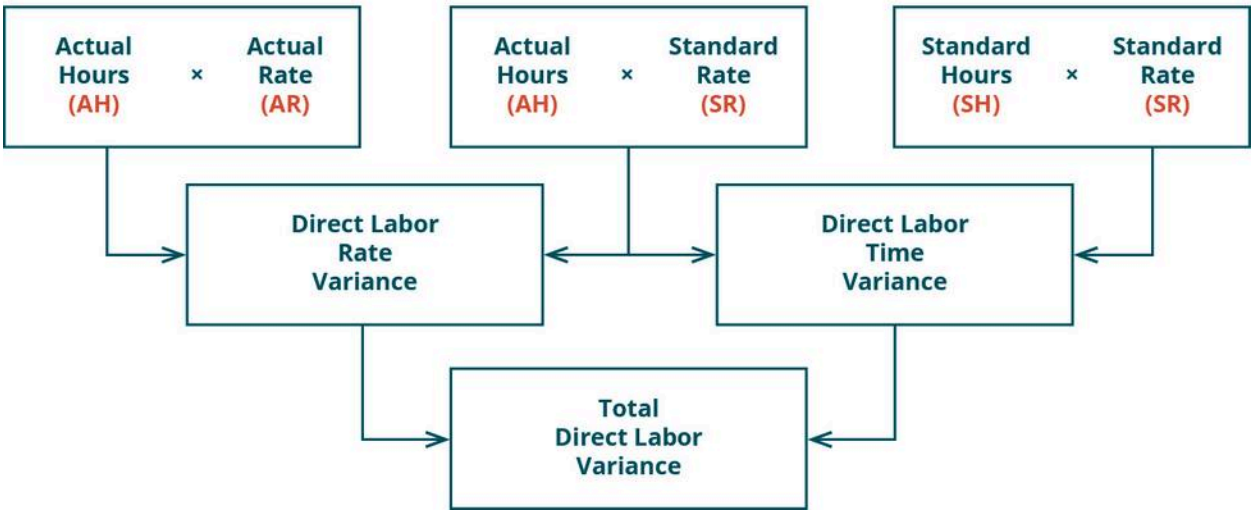


Figure 8.7 Direct Labor Variances. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

$$\text{Standard Overhead Rate} = \frac{\text{Budgeted Overhead Rate}}{\text{Level of Activity}}$$

$$\text{Variable Overhead Rate} = \frac{\text{Budgeted Variable Overhead}}{\text{Level of Activity}}$$

$$\text{Variable Overhead Rate Variance} = \left(\begin{array}{c}\text{Actual Hours Worked}\\ \times \\ \text{Actual Variable Overhead Rate per Hour}\end{array}\right) - \left(\begin{array}{c}\text{Actual Hours Worked}\\ \times \\ \text{Standard Variable Overhead Rate per Hour}\end{array}\right)$$

$$\text{Variable Overhead Rate Variance} = \left(\begin{array}{c}\text{Actual Variable}\\ \text{Overhead Rate}\end{array} - \begin{array}{c}\text{Standard Variable}\\ \text{Overhead Rate}\end{array}\right) \times \begin{array}{c}\text{Actual Hours}\\ \text{Worked}\end{array}$$

$$\text{Variable Overhead Efficiency Variance} = \left(\begin{array}{c}\text{Actual Hours Worked}\\ \times \\ \text{Standard Variable Overhead Rate per Hour}\end{array}\right) - \left(\begin{array}{c}\text{Standard Hours}\\ \times \\ \text{Standard Variable Overhead Rate per Hour}\end{array}\right)$$

$$\text{Variable Overhead Efficiency Variance} = \left(\text{Actual Labor Hours} - \text{Standard Labor Hours}\right) \times \begin{array}{c}\text{Standard}\\ \text{Overhead Rate}\end{array}$$

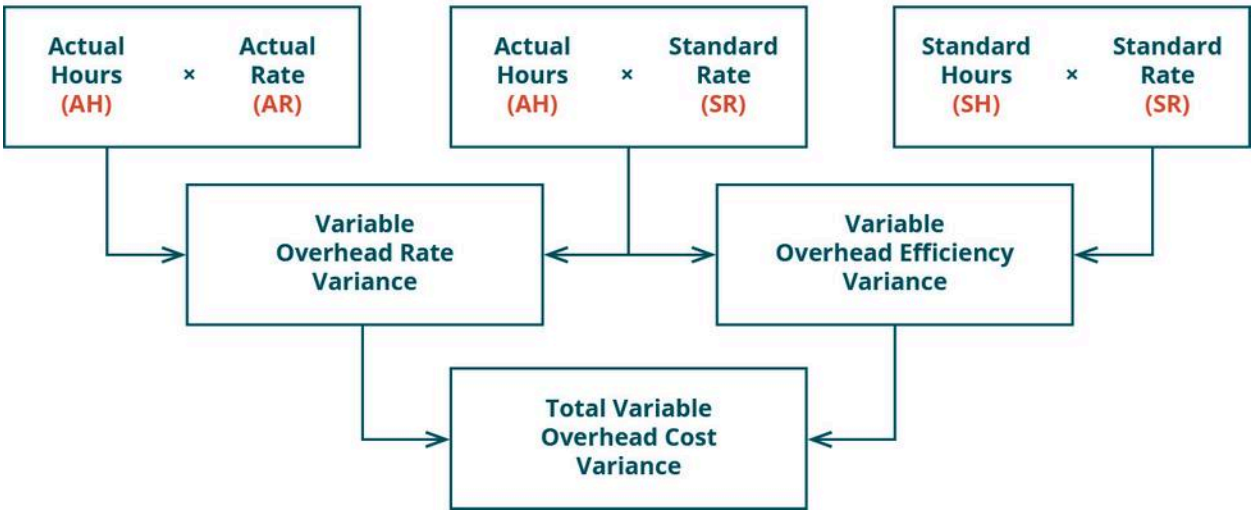


Figure 8.8 Overhead Variances. (attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license)

YOUR TURN

Barley, Inc. Production

Barley, Inc., produces a product and has the following as standard costs per unit for materials and labor:

Materials	4 pounds @ $15 per pound
Labor	2 hours @ $20 per hour

For the month of October, the following information was gathered related to production:

Beginning inventory	0
Units completed	10,000
Budgeted output units	12,000
Materials used (50,000 pounds)	$800,000
Labor (25,000 hours)	$450,000

Compute:

A. The materials price and quantity variances
B. The labor rate and efficiency variances

Provide possible explanations for each variance.

Solution

A.

Materials price variance:

$50,000 unfavorable = ($16* – $15) × 50,000 lb.

*$800,000/50,000

An unfavorable materials price variance occurred because the actual cost of materials was greater than the expected or standard cost. This could occur if a higher-quality material was purchased or the suppliers raised their prices.

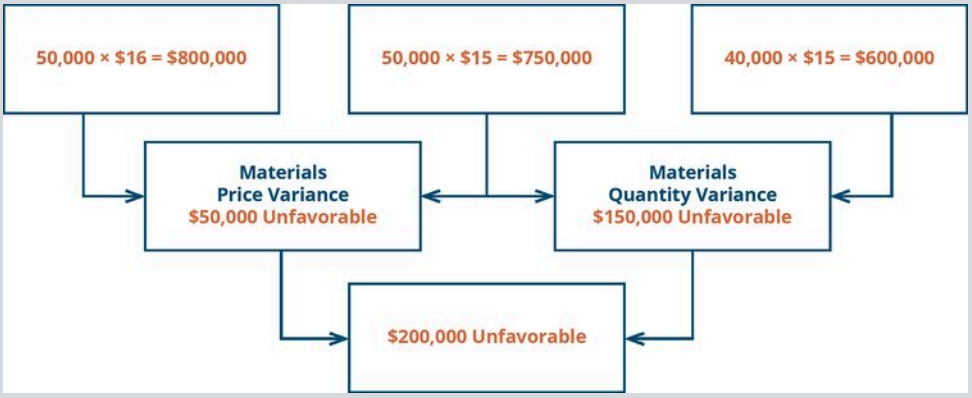
Materials quantity variance:

$150,000 unfavorable = (50,000 lb. – 40,000* lb.) × $15 per lb.

*4 lb. × 10,000 units

An unfavorable materials quantity variance occurred because the pounds of materials used were greater than the pounds expected to be used. This could occur if there were inefficiencies in production or the quality of the materials was such that more needed to be used to meet safety or other standards.

Materials inputs:



B.

Labor rate variance:

$50,000 favorable = ($18* per hour – $20 per hour) × 25,000 hours

*$450,000/25,000

A favorable labor rate variance occurred because the rate paid per hour was less than the rate expected to be paid (standard) per hour. This could occur because the company was able to hire workers at a lower rate, because of negotiated union contracts, or because of a poor labor rate estimate used in creating the standard.

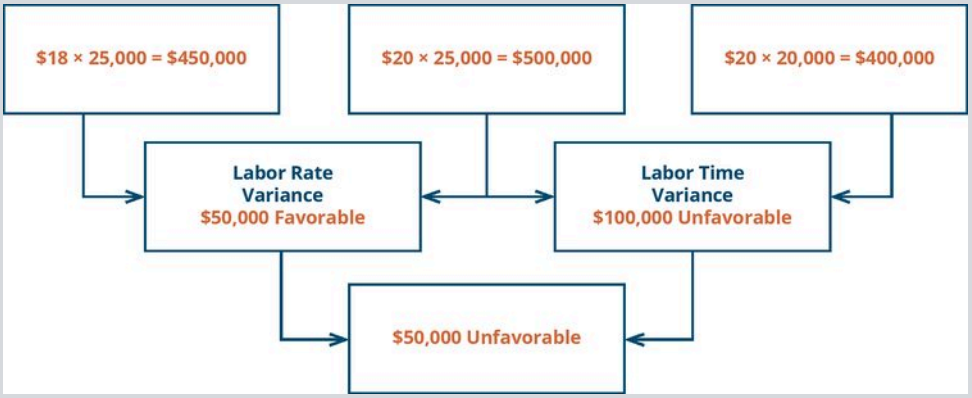
Labor quantity variance:

$100,000 unfavorable = (25,000 hours – 20,000* hours) × $20 per hour

*2 hours × 10,000 units

An unfavorable labor quantity variance occurred because the actual hours worked to make the 10,000 units were greater than the expected hours to make that many units. This could occur because of inefficiencies of the workers, defects and errors that caused additional time reworking items, or the use of new workers who were less efficient.

Labor inputs:



THINK IT THROUGH

Explaining Differences in Expected and Actual Operational Outcomes

The manager of a plant has called operations, purchasing, and personnel into her office to discuss the results of the last month. She notes that there was more than normal scrap, and employees worked more hours than expected. She is looking for an explanation for these results. What system might she have used to determine these material and labor issues? Why might these variances have occurred? What should she do about it for future periods?

LINK TO LEARNING

Standard Costing Advantages Explained

See this article on the four major advantages of standard costing (https://openstax.org/l/50StandCosting) to learn more.

Key Terms

attainable standard level that may be reached with reasonable effort
direct labor rate variance difference between the actual rate paid and the standard rate that should have been paid based on the actual hours worked
direct labor time variance difference between the actual hours worked and the standard hours that should have been worked for the actual units produced
direct labor variance measures how efficiently the company uses labor as well as how effective it is at pricing labor
direct materials price variance difference between the actual price paid per unit for materials and what should have been paid per the standards
direct materials quantity variance difference between the actual quantity of materials used and the standard materials that were expected to be used to make the actual units produced
direct materials variance difference between the actual price or amount used and the standard amount
favorable variance difference involving spending less, or using less, than the standard amount
fixed factory overhead variance difference between the actual fixed overhead and applied fixed overhead
flexible budget measurement and prediction of estimated revenues and costs at varying levels of production
ideal standard level that could be achieved if everything ran perfectly
standard expectation for a component used in production
standard cost cost expectation for price paid and amount (quantities) used
total direct labor variance actual labor costs compared to standard labor costs
total direct materials cost variance difference between actual materials cost and standard materials cost
total variable overhead cost variance total cost variance found by combining variable overhead rate variance and variable overhead efficiency variance
unfavorable variance difference involving spending more or using more than the standard amount
variable overhead efficiency variance difference between the actual hours worked and the standard hours expected for the units produced
variable overhead rate variance difference between the actual variable manufacturing overhead and the variable overhead that was expected given the number of hours worked
variance difference between standard and actual performance

Summary

8.1 Explain How and Why a Standard Cost Is Developed

- Standards are budgeted unit amounts for price paid and amount used.
- Variances are the difference between actual and standard amounts.
- A favorable variance is when the actual price or quantity is less than the standard amount.
- An unfavorable variance is when the actual price or amount is greater than the standard amount.

8.2 Compute and Evaluate Materials Variances

- There are two components to material variances: the direct materials price variance and the direct materials quantity variance.
- The direct materials price variance is caused by paying too much or too little for material.
- The direct materials quantity variance is caused by using too much or too little material.

8.3 Compute and Evaluate Labor Variances

- There are two labor variances: the direct labor rate variance and the direct labor time variance.
- The direct labor rate variance determines if the rate paid is greater than or less than the standard rate.
- The direct labor time variance determines if the actual hours used are greater than or less than the standards that should have been used.

8.4 Compute and Evaluate Overhead Variances

- There are two sets of overhead variances: variable and fixed.
- The variable variances are caused by the overhead application rate and the activity level against which the rate was applied.
- The variable overhead rate variance is the difference between the actual variable manufacturing overhead and the variable overhead that was expected given the number of hours worked.
- The variable overhead efficiency variance is driven by the difference between the actual hours worked and the standard hours expected for the units produced.
- There are two fixed overhead variances. One is caused by spending too much or too little on fixed overhead. The other is caused by actual production being above or below the expected production level.

8.5 Describe How Companies Use Variance Analysis

- The key to analyzing variances is to determine why the variance occurred.
- If a company cannot determine why there is a variance, it will not know if the variance is indicative of a problem or not.
- All firms—manufacturing, retail, and service—use standards and variances.

Multiple Choice

1. LO 8.1 Why does a company use a standard costing system?
A. to identify variances from actual cost that assist them in maintaining profits
B. to identify nonperformers in the workplace
C. to identify what vendors are unreliable
D. to identify defective materials

2. LO 8.1 This standard is set at a level that may be reached with reasonable effort.
A. ideal standard
B. attainable standard
C. unattainable standard
D. variance from standard

3. LO 8.1 This standard is set at a level that could be achieved if everything ran perfectly.
A. ideal standard
B. attainable standard
C. unattainable standard
D. variance from standard

4. LO 8.1 This variance is the difference involving spending more or using more than the standard amount.
A. favorable variance
B. unfavorable variance
C. no variance
D. variance

5. LO 8.1 This variance is the difference involving spending less, or using less than the standard amount.
A. favorable variance
B. unfavorable variance
C. no variance
D. variance

6. LO 8.2 What are some possible reasons for a material price variance?

A. substandard material
B. labor rate increases
C. labor rate decreases
D. labor efficiency

7. LO 8.2 When is the material price variance unfavorable?
A. when the actual quantity used is greater than the standard quantity
B. when the actual quantity used is less than the standard quantity
C. when the actual price paid is greater than the standard price
D. when the actual price is less than the standard price

8. LO 8.2 When is the material price variance favorable?
A. when the actual quantity used is greater than the standard quantity
B. when the actual quantity used is less than the standard quantity
C. when the actual price paid is greater than the standard price
D. when the actual price is less than the standard price

9. LO 8.2 What are some reasons for a material quantity variance?
A. building rental charges increase
B. labor rate decreases
C. more qualified workers
D. change in the actual cost of materials

10. LO 8.2 When is the material quantity variance favorable?
A. when the actual quantity used is greater than the standard quantity
B. when the actual quantity used is less than the standard quantity
C. when the actual price paid is greater than the standard price
D. when the actual price is less than the standard price

11. LO 8.2 When is the material quantity unfavorable?
A. when the actual quantity used is greater than the standard quantity
B. when the actual quantity used is less than the standard quantity
C. when the actual price paid is greater than the standard price
D. when the actual price is less than the standard price

12. LO 8.3 What are some possible reasons for a labor rate variance?
A. hiring of less qualified workers
B. an excess of material usage
C. material price increase
D. utilities usage change

13. LO 8.3 When is the labor rate variance unfavorable?
A. when the actual quantity used is greater than the standard quantity
B. when the actual quantity used is less than the standard quantity
C. when the actual price paid is greater than the standard price
D. when the actual price is less than the standard price

14. LO 8.3 When is the labor rate variance favorable?
A. when the actual quantity used is greater than the standard quantity
B. when the actual quantity used is less than the standard quantity

C. when the actual price is greater than the standard price
D. when the actual price is less than the standard price

15. LO 8.3 What are some possible reasons for a direct labor time variance?
A. utility usage decrease
B. less qualified workers
C. office supplies spending
D. sales decline

16. LO 8.3 When is the direct labor time variance favorable?
A. when the actual quantity used is greater than the standard quantity
B. when the actual quantity used is less than the standard quantity
C. when the actual price paid is greater than the standard price
D. when the actual price is less than the standard price

17. LO 8.3 When is the direct labor time variance unfavorable?
A. when the actual quantity used is greater than the standard quantity
B. when the actual quantity used is less than the standard quantity
C. when the actual price paid is greater than the standard price
D. when the actual price is less than the standard price

18. LO 8.4 A flexible budget _____.
A. predicts estimated revenues and costs at varying levels of production
B. gives actual figures for selling price
C. gives actual figures for variable and fixed overhead
D. is not used in overhead variance calculations

19. LO 8.4 The variable overhead rate variance is caused by the difference between which of the following?
A. actual and standard allocation base
B. actual and standard overhead rates
C. actual and budgeted units
D. actual units and actual overhead rates

20. LO 8.4 The variable overhead efficiency variance is caused by the difference between which of the following?
A. actual and budgeted units
B. actual and standard allocation base
C. actual and standard overhead rates
D. actual units and actual overhead rates

21. LO 8.4 The fixed factory overhead variance is caused by the difference between which of the following?
A. actual and standard allocation base
B. actual and budgeted units
C. actual fixed overhead and applied fixed overhead
D. actual and standard overhead rates

22. LO 8.5 Which of the following is a possible cause of an unfavorable material price variance?
A. purchasing too much material
B. purchasing higher-quality material
C. hiring substandard workers

D. buying substandard material

23. LO 8.5 Which of the following is a possible cause of an unfavorable material quantity variance?
A. purchasing substandard material
B. hiring higher-quality workers
C. paying more than should have for workers
D. purchasing too much material

24. LO 8.5 Which of the following is a possible cause of an unfavorable labor efficiency variance?
A. hiring substandard workers
B. making too many units
C. buying higher-quality material
D. paying too much for workers

25. LO 8.5 Which of the following is a possible cause of an unfavorable labor rate variance?
A. hiring too many workers
B. hiring higher-quality workers at a higher wage
C. making too many units
D. purchasing too much material

Questions

1. LO 8.1 What two components are needed to determine a standard for materials?

2. LO 8.1 What two components are needed to determine a standard for labor?

3. LO 8.1 What elements require consideration before establishing an overhead standard?

4. LO 8.1 What is a variance?

5. LO 8.2 What causes the material price variance?

6. LO 8.2 What causes the material quantity variance?

7. LO 8.2 What are some possible causes of a material price variance?

8. LO 8.2 What are some possible causes of a material quantity variance?

9. LO 8.3 What is the direct labor rate variance?

10. LO 8.3 What is the direct labor time variance?

11. LO 8.3 What are some possible causes of a direct labor rate variance?

12. LO 8.3 What are some possible causes of a direct labor time variance?

13. LO 8.3 How is the total direct labor variance calculated?

14. LO 8.4 What causes the variable overhead rate variance?

15. LO 8.4 What causes the variable overhead efficiency variance?

16. LO 8.4 What is the main difference between a flexible budget and a master budget?

17. LO 8.5 What causes a favorable variance?

18. LO 8.5 What causes an unfavorable variance?

19. LO 8.5 When might a favorable variance not be a good outcome?

20. LO 8.5 When might an unfavorable variance be a good outcome?

21. LO 8.5 Identify several causes of a favorable material price variance.

22. LO 8.5 Identify several causes of an unfavorable material price variance.

23. LO 8.5 Identify several causes of a favorable material quantity variance.

24. LO 8.5 Identify several causes of an unfavorable material quantity.

25. LO 8.5 Identify several causes of a favorable labor rate variance.

26. LO 8.5 Identify several causes of an unfavorable labor rate variance.

27. LO 8.5 Identify several causes of a favorable labor efficiency variance.

28. LO 8.5 Identify several causes of an unfavorable labor efficiency variance.

Exercise Set A

EA 1. LO 8.1 Moisha is developing material standards for her company. The operations manager wants grade A widgets because they are the easiest to work with and are the quality the customers want. Grade B will not work because customers do not want the lower grade, and it takes more time to assemble the product than with grade A materials. Moisha calls several suppliers to get prices for the widget. All are within $0.05 of each other. Since they will use millions of widgets, she decides that the $0.05 difference is important. The supplier who has the lowest price is known for delivering late and low-quality materials. Moisha decides to use the supplier who is $0.02 more but delivers on time and at the right quality. This supplier charges $0.48 per widget. Each unit of product requires four widgets. What is the standard cost per unit for widgets?

EA 2. LO 8.1 Rene is working with the operations manager to determine what the standard labor cost is for a spice chest. He has watched the process from start to finish and taken detailed notes on what each employee does. The first employee selects and mills the wood, so it is smooth on all four sides. This takes the employee 1 hour for each chest. The next employee takes the wood and cuts it to the proper size. This takes 30 minutes. The next employee assembles and sands the chest. Assembly takes 2 hours. The chest then goes to the finishing department. It takes 1.5 hours to finish the chest. All employees are cross-trained so they are all paid the same amount per hour, $17.50.

A. What are the standard hours per chest?
B. What is the standard cost per chest for labor?

EA 3. LO 8.1 Fiona cleans offices. She is allowed 5 seconds per square foot. She cleans building A, which is 3,000 square feet, and building B, which is 2,460 square feet. Will she finish these two buildings in an 8-hour shift? Will she have time for a break?

EA 4. LO 8.1 Use the information provided to create a standard cost card for production of one glove box switch. To make one switch it takes 16 feet of plastic-coated copper wire and 0.5 pounds of plastic material. The plastic material can usually be purchased for $20.00 per pound, and the wire costs $2.50 per foot. The labor necessary to assemble a switch consists of two types. The first type of labor is assembly, which takes 3.5 hours. These workers are paid $27.00 per hour. The second type of labor is finishing, which takes 2 hours. These workers are paid $29.00 per hour. Overhead is applied using labor hours. The variable overhead rate is $14.90 per labor hour. The fixed overhead rate is $15.60 per hour.

EA 5. LO 8.2 Sitka Industries uses a cost system that carries direct materials inventory at a standard cost. The controller has established these standards for one ladder (unit):

	Standard Quantity	×	Standard Price	=	Standard Cost
Direct materials	3 pounds		$4.50 per pound		$13.50
Direct labor	2.00 hours		$12.00 per hour		24.00
Total cost					$37.50

Sitka Industries made 3,000 ladders in July and used 8,800 pounds of material to make these units. Sitka Industries bought 15,500 pounds of material in the current period. There was a $250 unfavorable direct materials price variance.

A. How much in total did Sitka pay for the 15,500 pounds?
B. What is the direct materials quantity variance?
C. What is the total direct material cost variance?
D. What if 9,500 pounds were used to make these ladders, what would be the direct materials quantity variance?
E. If there was a $340 favorable direct materials price variance, how much did Sitka pay for the 15,500 pounds of material?

EA 6. LO 8.2 Use the information provided to answer the questions.

Actual price paid per pound of material	$ 14.50
Total standard pounds for units produced this period	12,500
Pounds of material used	13,250
Direct materials price variance favorable	$4,637.50

All material purchased was used in production.

A. What is the standard price for materials?
B. What is the direct materials quantity variance?
C. What is the total direct materials cost variance?
D. If the direct materials price variance was unfavorable, what would be the standard price?

EA 7. LO 8.2 Dog Bone Bakery, which bakes dog treats, makes a special biscuit for dogs. Each biscuit uses 0.75 cup of pure semolina flour. They buy 4,000 cups of flour at $0.55 per cup. They use 3,550 cups of flour to make 4,750 biscuits. The standard cost per cup of flour is $0.53.

A. What are the direct materials price variance, the direct materials quantity variances, and the total direct materials cost variance?
B. What is the standard cost per biscuit for the semolina flour?

EA 8. LO 8.3 Queen Industries uses a standard costing system in the manufacturing of its single product. It requires 2 hours of labor to produce 1 unit of final product. In February, Queen Industries produced 12,000 units. The standard cost for labor allowed for the output was $90,000, and there was an unfavorable direct labor time variance of $5,520.

A. What was the standard cost per hour?
B. How many actual hours were worked?
C. If the workers were paid $3.90 per hour, what was the direct labor rate variance?

EA 9. LO 8.3 Penny Company manufactures only one product and uses a standard cost system. The following information is from Penny's records for May:

Direct labor rate variance	$15,000 favorable
Direct labor time variance	$25,000 unfavorable
Standard hours per unit produced	2.5
Standard rate per hour	$25

During May, the company used 12.5% more hours than the standard allowed.

A. What were the total standard hours allowed for the units manufactured during the month?
B. What were the actual hours worked?
C. How many actual units were produced during May?

EA10. LO 8.4 ThingOne Company has the following information available for the past year. They use machine hours to allocate overhead.

Actual total overhead	$75,000
Actual fixed overhead	$32,500
Actual machine hours	10,000
Standard hours for the units produced	9,500
Standard variable overhead rate	$ 4.50

What is the variable overhead efficiency variance?

EA11. LO 8.4 A manufacturer planned to use $78 of variable overhead per unit produced, but in the most recent period, it actually used $76 of variable overhead per unit produced. During this same period, the company planned to produce 500 units but actually produced 540 units. What is the variable overhead spending variance?

EA12. LO 8.5 Acme Inc. has the following information available:

Actual price paid for material	$1.00
Standard price for material	$1.20
Actual quantity purchased and used in production	100
Standard quantity for units produced	110
Actual labor rate per hour	$ 15
Standard labor rate per hour	$ 16
Actual hours	200
Standard hours for units produced	220

A. Compute the material price and quantity, and the labor rate and efficiency variances.
B. Describe the possible causes for this combination of favorable and unfavorable variances.

EA13. LO 8.5 Acme Inc. has the following information available:

Actual price paid for material	$1.00
Standard price for material	$0.90
Actual quantity purchased and used in production	100
Standard quantity for units produced	110
Actual labor rate per hour	$ 15
Standard labor rate per hour	$ 16
Actual hours	200
Standard hours for units produced	220

A. Compute the material price and quantity, and the labor rate and efficiency variances.
B. Describe the possible causes for this combination of favorable and unfavorable variances.

EA14. LO 8.5 Acme Inc. has the following information available:

Actual price paid for material	$1.00
Standard price for material	$0.90
Actual quantity purchased and used in production	100
Standard quantity for units produced	110
Actual labor rate per hour	$ 15
Standard labor rate per hour	$ 14
Actual hours	200
Standard hours for units produced	220

A. Compute the material price and quantity, and the labor rate and efficiency variances.
B. Describe the possible causes for this combination of favorable and unfavorable variances.

Exercise Set B

EB 1. LO 8.1 Bristol is developing material standards for her company. The operations manager wants grade A plastic tops because they are the easiest to work with and are the quality the customers want. Grade B will not work because customers do not want the lower grade, and it takes more time to assemble the product than with grade A materials. Bristol calls several suppliers to get prices for the plastic top. All are within $0.10 of each other. Since the company will use millions of the plastic tops, she decides that the $0.10 difference is important. The supplier who has the lowest price is known for delivering late and low-quality materials. Bristol decides to use the supplier who is $0.04 more but delivers on time and at the right quality. This supplier charges $0.52 per plastic top. Each unit of product requires six plastic tops. What is the standard cost per unit for plastic tops?

EB 2. LO 8.1 Salley is developing material and labor standards for her company. She finds that it costs $0.55 per pound of material per widget. Each widget requires 6 pounds of material per widget. Salley is also working with the operations manager to determine what the standard labor cost is for a widget. Upon observation, Salley notes that it takes 3 hours in the assembly department and 1 hour in the finishing department to complete one widget. All employees are paid $10.50 per hour.
A. What is the standard materials cost per unit for a widget?
B. What is the standard labor cost per unit for a widget?

EB 3. LO 8.1 Use the following information to create a standard cost card for production of one photography drone from Drone Experts.
To make one drone it takes 2 pounds of plastic material. The material can usually be purchased for $25.00 per pound. The labor necessary to build a drone consists of two types. The first type of labor is assembly, which takes 10.5 hours. These workers are paid $21.00 per hour. The second type of labor is finishing, which takes 7 hours. These workers are paid $25.00 per hour. Overhead is applied using labor hours. The variable overhead rate is $14.00 per labor hour. The fixed overhead rate is $16.00 per hour.

EB 4. LO 8.1 Mateo makes gizmos. He would like to set up a system to help him manage his business. The gizmos are made in a standard process. There is a certain amount of material and labor that goes into each gizmo. The only difference between the gizmo is the color of the material. What information should Mateo collect, how should he format it, and what kind of reports should he prepare to help him run his business?

EB 5. LO 8.2 Smith Industries uses a cost system that carries direct materials inventory at a standard cost. The controller has established these standards for the cost of one basket (unit):

	Standard Quantity	×	Standard Price	=	Standard Cost
Direct materials	5 pounds		$2.60 per pound		$18.00
Direct labor	1.25 hours		$12.00 per hour		$15.00
Total prime cost					$33.00

Smith Industries made 3,000 baskets in July and used 15,500 pounds of material to make these units. Smith Industries paid $39,370 for the 15,500 pounds of material.

A. What was the direct materials price variance for July?
B. What was the direct materials quantity variance for July?
C. What is the total direct materials cost variance?
D. If Smith Industries used 15,750 pounds to make the baskets, what would be the direct materials quantity variance?

EB 6. LO 8.2 Lizbeth, Inc., makes ice cream. The toffee coffee ice cream takes 4 quarts of cream, 3 cups of sugar, 2 tablespoons of toffee flavoring, and 1.5 tablespoons of coffee flavoring per gallon. The standard prices are $2.00 per quart of cream, $0.40 per cup of sugar, $0.50 per tablespoon of toffee flavoring, and $0.75 per tablespoon of coffee flavoring.

A. What is the standard material cost for a gallon of toffee coffee ice cream?
B. If Lizbeth makes 35 gallons of toffee coffee ice cream, how much of each of the ingredients should she use?
C. If Lizbeth uses 105 quarts of cream to make 25 gallons of ice cream, what would be the cream (direct materials) quantity variance?
D. If Lizbeth uses 45 tablespoons of toffee flavoring to make 25 gallons of ice cream, what would be the toffee flavoring (direct materials) quantity variance?

EB 7. LO 8.2 Woodpecker manufactures sawmill equipment. They use a standard costing system and recognize material price variance at the time of material purchases. They use carbide to make the teeth on their band-saw blades. They received an order for 250 band-saw blades, but they did not have any carbide in stock. They purchased 3,500 pounds of carbide for $14,875 but should have spent $16,275. Each saw blade has a standard carbide direct materials quantity of 7.8 pounds.

A. If they used 8 pounds per blade, what would be the direct materials quantity variance?
B. If they used 7.5 pounds per blade, what would be the direct materials quantity variance?
C. Compute the direct materials price variance based on 7.5 pounds of carbide per blade actually used.

EB 8. LO 8.3 Case made 24,500 units during June, using 32,000 direct labor hours. They expected to use 31,450 hours per the standard cost card. Their employees were paid $15.75 per hour for the month of June. The standard cost card uses $15.50 as the standard hourly rate.

A. Compute the direct labor rate and time variances for the month of June, and also calculate the total direct labor variance.
B. If the standard rate per hour was $16.00, what would change?

EB 9. LO 8.3 Eagle Inc. uses a standard cost system. During the most recent period, the company manufactured 115,000 units. The standard cost sheet indicates that the standard direct labor cost per unit is $1.50. The performance report for the period includes an unfavorable direct labor rate variance of $3,700 and a favorable direct labor time variance of $10,275.
What was the total actual cost of direct labor incurred during the period?

EB10. LO 8.4 A manufacturer planned to use $45 of variable overhead per unit produced, but in the most

recent period, it actually used $47 of variable overhead per unit produced. During this same period, the company planned to produce 200 units but actually produced 220 units. What is the variable overhead spending variance?

EB11. LO 8.4 Fitzgerald Company manufactures sewing machines, and they produced 2,500 this past month. The standard variable manufacturing overhead (MOH) rate used by the company is $6.75 per machine hour. Each sewing machine requires 13.5 machine hours. Actual machine hours used last month were 33,500, and the actual variable MOH rate last month was $7.00.
Calculate the variable overhead rate variance and the variable overhead efficiency variance.

EB12. LO 8.5 Acme Inc. has the following information available:

Actual price paid for material	$1.00
Standard price for material	$0.90
Actual quantity purchased and used in production	100
Standard quantity for units produced	90
Actual labor rate per hour	$ 15
Standard labor rate per hour	$ 14
Actual hours	200
Standard hours for units produced	190

A. Compute the material price and quantity, and the labor rate and efficiency variances.
B. Describe the possible causes for this combination of favorable and unfavorable variances.

EB13. LO 8.5 Acme Inc. has the following information available:

Actual price paid for material	$1.00
Standard price for material	$1.10
Actual quantity purchased and used in production	100
Standard quantity for units produced	110
Actual labor rate per hour	$ 15
Standard labor rate per hour	$ 14
Actual hours	200
Standard hours for units produced	190

A. Compute the material price and quantity, and the labor rate and efficiency variances.
B. Describe the possible causes for this combination of favorable and unfavorable variances.

EB14. LO 8.5 Acme Inc. has the following information available:

Actual price paid for material	$1.00
Standard price for material	$0.90
Actual quantity purchased and used in production	100
Standard quantity for units produced	90
Actual labor rate per hour	$ 15
Standard labor rate per hour	$ 16
Actual hours	200
Standard hours for units produced	220

A. Compute the material price and quantity, and the labor rate and efficiency variances.
B. Describe the possible causes for this combination of favorable and unfavorable variances.

Problem Set A

PA 1. LO 8.1 The comptroller wants to set the standards according to a study done by a consulting firm for a company. The consulting firm used the following assumptions: The machines never break down.

Workers never take a break. The material used is perfect. The material arrives on time. No one takes a day off. Workers are well trained. Workers do not make defective units. What kinds of standards are these? Will the workers be motivated to achieve these standards?

PA 2. LO 8.1 Stan is opening a coffee shop next to Big State University. He knows that controlling his costs will be important to the success of the shop. He will not be able to work all the hours the shop is open, so the employees will need some guidelines to perform their jobs correctly. After talking to an accounting professor, he decides he needs a standard cost system for his shop. Describe the process Stan should follow in setting his standards for materials and labor.

PA 3. LO 8.1 What makes a variance favorable? Give an example of a favorable variance involving materials. What makes a variance unfavorable? Give an example of an unfavorable variance involving labor.

PA 4. LO 8.2 April Industries employs a standard costing system in the manufacturing of its sole product, a park bench. They purchased 60,000 feet of raw material for $300,000, and it takes 5 feet of raw materials to produce one park bench. In August, the company produced 10,000 park benches. The standard cost for material output was $100,000, and there was an unfavorable direct materials quantity variance of $6,000.

A. What is April Industries' standard price for one unit of material?
B. What was the total number of units of material used to produce the August output?
C. What was the direct materials price variance for August?

PA 5. LO 8.2 Ed Co. manufactures two types of O rings, large and small. Both rings use the same material but require different amounts. Standard materials for both are shown.

	Large	Small
Rubber	3 feet at $0.25 per foot	1.25 feet at $0.25 per foot
Connector	1 at $0.03	1 at $0.03

At the beginning of the month, Ed Co. bought 25,000 feet of rubber for $6,875. The company made 3,000 large O rings and 4,000 small O rings. The company used 14,500 feet of rubber.

A. What are the direct materials price variance, the direct materials quantity variance, and the total direct materials cost variance?
B. If they bought 10,000 connectors costing $310, what would the direct materials price variance be for the connectors?
C. If there was an unfavorable direct materials price variance of $125, how much did they pay per foot for the rubber?

PA 6. LO 8.2 The Whizbang Company makes a special type of toy. Each toy takes 6 ounces of a special material that costs $3 per ounce. Whizbang bought 4,000 ounces of the material at a cost of $11,300. They used 3,400 ounces to make 534 toys. Compute the direct materials price variance, the direct materials quantity variance, and the total direct materials quantity variance.

PA 7. LO 8.3 Ellis Company's labor information for September is as follows:

Direct labor hourly rate paid	$ 32.00
Total standard direct labor hours for units produced	11,000
Direct labor hours worked	10,850
Direct labor rate variance	$8,137.50 (favorable)

A. Compute the standard direct labor rate per hour.
B. Compute the direct labor time variance.
C. Compute the standard direct labor rate if the direct labor rate variance was $2,712.50 (unfavorable).

PA 8. LO 8.3 Breakaway Company's labor information for May is as follows:

Actual direct labor hours worked	48,000
Standard direct labor hours allowed	47,400
Total payroll for direct labor	$1,128,000
Direct labor time variance	$ 13,800 (unfavorable)

A. What is the actual direct labor rate per hour?
B. What is the standard direct labor rate per hour?
C. What was the total standard direct labor cost for May?
D. What was the direct labor rate variance for May?

PA 9. LO 8.3 Power Co.'s labor information for June is as follows:

Direct labor hours worked	67,200
Standard labor hours for units manufactured	70,000
Unfavorable direct labor rate variance	$ 23,520
Total payroll for labor	$823,200

A. What was the actual labor rate per hour?
B. What was the standard labor rate per hour?
C. What was the total standard labor cost for units produced in June?
D. What was the direct labor time variance for June?

PA10. LO 8.4 Prepare a flexible budget for overhead based on the following data:

Percent of capacity	**90%**	**100%**	**110%**
Direct labor hours	3,600	4,000	4,400
Units of output	900	1,000	1,100
Variable overhead	$3,600	$ 4,000	$ 4,400
Fixed overhead	$6,000	$ 6,000	$ 6,000
Total overhead	$9,600	$10,000	$10,400

Normal capacity = 100% and overhead is applied based on direct labor hours
Standard overhead rate = $10,000/4,000 = $2.50 per direct labor hour
Direct materials are $67.50 per unit.
Direct labor is $23.50 per hour.

PA11. LO 8.4 Reddy Corporation has collected the following data for the month of June:

Actual total factory overhead incurred	$61,250
Budgeted fixed factory overhead costs	$42,000
Activity level, in direct labor hours	15,000
Actual direct labor hours	18,000
Standard hours for output this period	17,000
Total factory overhead rate	$ 4.35

What is the variable overhead efficiency variance?

PA12. LO 8.4 ABC Inc. spent a total of $48,000 on factory overhead. Of this, $28,000 was fixed overhead. ABC

Inc. had budgeted $27,000 for fixed overhead. Actual machine hours were 5,000. Standard hours for units made were 4,800. The standard variable overhead rate was $4.10. What is the variable overhead rate variance?

PA13. LO 8.5 Recompute the variances from the second Acme Inc. exercise using $0.0725 as the standard cost of the material and $14 as the standard labor cost per hour. How has your explanation of the variances changed?

Problem Set B

PB 1. LO 8.1 Sameerah is trying to determine the standard hours to make one unit. She has studied the manufacturing process and is trying to determine what portion of the employees' time should be included in the standard time to make the product. She knows that the actual time the worker is assembling, cutting, and painting should be part of the standard hours. She is questioning whether setup, down time, rest periods, and cleanup should be part of the standard hours. Explain why you would or would not include these times.

PB 2. LO 8.1 Carl cleans offices. He has the following buildings to clean every day: building A, which is 12,500 square feet; building B, which is 24,500 square feet; building C, which is 10,500 square feet; and building D, which is 6,700 square feet. He is allowed 5 seconds per square foot. Each employee is allowed one 30-minute lunch per shift. How many employees will he need to hire?

PB 3. LO 8.1 Freidrich is working with the operations manager to determine what the standard material cost is for a spice chest. He has watched the process from start to finish and taken detailed notes on what material is used. The easiest material to measure is the wood. Each chest uses 5 board feet and produces 1.5 feet of scrap. He is not sure what to do with the scrap that is produced; the company cannot buy the boards in any other dimensions. What amount of materials should be included in the standard for material costs?

PB 4. LO 8.2 A company bought 45,000 pounds of plastic pellets to make DVDs at a cost of $9,900. The standard cost per pound for the pellets is 20.5 cents. Some of these pellets were used in three jobs. The first job called for 7,500 pounds but used 7,250 pounds. The second job called for 8,800 pounds but used 9,000 pounds. The third job called for 2,300 pounds but used 2,250 pounds. Compute the direct materials price variance and the direct materials quantity variance for each job and in total. Why would you want to calculate the direct materials quantity variance for each job?

PB 5. LO 8.2 Illinois Company is a medium-sized company that makes dresses. During the month of June, 8,575 dresses were made. All material purchases were used to make the dresses. The company had this information: standard per dress of 6 yards of material at $6.20 per yard. The actual quantity was 52,000 yards at a cost of $325,520. Compute the direct materials price variance, the direct materials quantity variance, and the total direct materials cost variance.

PB 6. LO 8.2 Corolla Manufacturing has a standard cost for steel of $20 per pound for a product that uses 4 pounds of steel. During September, Corolla purchased and used 4,200 pounds of steel to make 1,040 units. They paid $20.75 per pound for the steel. Compute the direct materials price variance, the direct materials quantity variance, and the total direct materials cost variance for the month of September. What would change if Corolla had made 2,200 units?

PB 7. LO 8.3 Marymount Company makes one product. In the month of April, it made 3,500 units. Workers were paid $32 per hour for labor, for a total of $718,848. The standard hours per unit are 6.4, and the standard labor wage rate is $38.40 per hour.

A. What are the actual hours worked?
B. What are the standard hours for the units made?
C. What is the direct labor rate variance for April?

D. What is the direct labor time variance for April?
E. What is the total direct labor variance for April?

PB 8. LO 8.3 Adam Inc.'s records for May include the following information:

Actual payroll	$40,000
Actual hours worked	3,200
Labor rate variance	$ 8,000 (favorable)
Labor time variance	$ 4,800 (unfavorable)

A. What are Adam's standard labor hours for the units made?
B. What is Adam's total standard labor cost for the units made?

PB 9. LO 8.3 Ribco's labor cost information for making its only product for March is as follows:

Standard hours per unit	1.25
Budgeted units for the period	12,000
Finished units for the period	10,000
Standard rate per hour	$ 40
Actual labor costs incurred	$414,000
Actual rate paid per hour	$ 36

A. What is the direct labor rate variance?
B. What is the direct labor time variance?
C. What is the total direct labor variance?

PB10. LO 8.5 Use the following standard cost card for 1 gallon of ice cream to answer the questions.

STANDARD COST CARD

Product: Gallon of Ice Cream

Manufacturing Cost Information	Standard Quantity	×	Standard Cost per Unit	=	Cost Summary
Direct Materials					
Cream	5 quarts		$1.15 per quart		$5.75
Sugar	16 ounces		$0.08 per ounce		$1.28
Direct Labor	3 minutes		$36.00 per hour		$1.80
Total Direct Costs					$8.83

Actual direct costs incurred to make 50 gallons of ice cream:

- 275 quarts of cream at $1.05 per quart
- 832 ounces of sugar at $0.075 per ounce
- 165 minutes of labor at $37 per hour

All material used was bought during the current period.

A. Compute the material and labor variances.
B. Comment on the results and possible causes of the variances.

PB11. LO 8.5 Use the following standard cost card for 1 gallon of ice cream to answer the questions.

STANDARD COST CARD Product: Gallon of Ice Cream				
Manufacturing Cost Information	**Standard Quantity**	**×**	**Standard Cost per Unit**	**=**	**Cost Summary**
Direct Materials				
Cream	6 quarts		$1.00 per quart		$6.00
Sugar	15 ounces		$0.07 per ounce		$1.05
Direct Labor	3 minutes		$38.00 per hour		$1.90
Total Direct Costs					$8.95

Actual direct costs incurred to make 50 gallons of ice cream:

- 275 quarts of cream at $1.05 per quart
- 832 ounces of sugar at $0.075 per ounce
- 165 minutes of labor at $37 per hour

All materials used were bought during the current period.

A. Compute the material and labor variances.
B. Comment on the results and possible causes of the variances.

Thought Provokers

TP 1. LO 8.1 How do you balance a firm's need to succeed and the need for not asking the workers for perfection?

TP 2. LO 8.1 What type of firm would use standard costing? What type of firm would not use standard costing?

TP 3. LO 8.2 You started your own construction business and need to determine the cost of materials used to build one house, and how many materials you will need to do so.

- Where would you begin to determine the standard price and quantity needs to build one house?
- What would produce a difference between the standard cost to build a house and the actual cost? What would cause a favorable outcome? What would cause an unfavorable outcome?
- What action might your company take if you had an unfavorable total direct materials cost variance?

TP 4. LO 8.3 Is labor a true variable cost?

TP 5. LO 8.4 Why would managers use a flexible budget? What information does it provide that a regular budget does not?

TP 6. LO 8.4 Fill in the blanks in the following flexible budget:

Percent of Capacity		**100%**	
Direct labor hours	3,600		
Units of output	1,800	2,000	2,200
Variable overhead		$ 8,000	
Fixed overhead		$10,000	
Total overhead		$18,000	

TP 7. LO 8.4 Before automation became more prevalent, overhead was often calculated and allocated as a function of direct labor costs or direct labor hours. Why was this the case, and has this pattern changed?

TP 8. LO 8.5 In your opinion, is it important that an organization set standards and measure them monthly?

Why or why not?

Chapter 7

Multiple Choice

1. D
3. A
5. B
7. D
9. C
11. C
13. C
15. C
17. C

Questions

1. A budget is a written financial plan for a set period, which is typically a year. There are several different types of budgets including the master budget, operating budget, financial budget, flexible budget, and operating budget.
3. This approach begins at the lowest levels of management. These managers know the details involved with their departments. This allows for more accurate budget estimates when management understands how their department contributes to the company's goals. Disadvantages include that this type of budgeting takes time, which leads to more labor costs, and when management doesn't fully understand how it contributes to the company goals, the budget may support the department and not the company.
5. Operating budgets plan the primary operations of the business and need accurate information in order to provide accurate planning. Assumptions such as sales in units, sales price, desired ending inventory in units, manufacturing costs per unit, which include direct material needed per unit, desired direct materials ending inventory, amount of direct labor hours and rate, and the overhead required for production and managing the company.
7. The budgeted income statement includes the estimated revenue and expenses for the company. Using historical data on cash collections helps plan when the cash will be received and is used to develop the cash collections schedule. The company applies its payment policies on its purchases and other items requiring cash expenditures. This creates the cash payments schedule. Information from the cash collections schedule, cash payments schedule, and the capital expense budget are combined to develop the cash budget. The information from the cash budget and the ending balance sheet from the preceding year are used to develop the budgeted balance sheet.
9. A. cash budget; B. cash receipts budget; C. production budget; D. cash payments schedule; E. capital assets budget
11. This budget is the plan for the purchase and disposal of plant assets and lists the estimated dollar amounts for each.
13. Before the time period begins, the organization's goals should be defined so the budget can be set to achieve the goals. During the time period, the results should be properly measured and reported so necessary changes can be made during the year. Then, the results of the operations can be evaluated and compared to the original budget and organization's goals.

Chapter 8

Multiple Choice

1. A
3. A
5. A
7. C

9. C
11. A
13. C
15. B
17. A
19. B
21. C
23. A
25. B

Questions

1. The expected price of materials per unit and the expected quantity usage are needed to help determine a standard.
3. Fixed overhead and variable overhead should be considered.
5. Paying more or less than the standard price
7. Buying a different quality level of material; good or bad purchasing/negotiation
9. A direct labor rate variance is the actual rate paid being different from the standard rate.
11. Employees have a different level of experience than standards; the labor market is tighter or looser than expected; contract renegotiation.
13. Total direct labor variance = (Actual hours × Actual rate) – (Standard hours × Standard rate) or the total direct labor variance is also found by combining the direct labor rate variance and the direct labor time variance.
15. The difference between the actual and standard amounts of the allocation base cause variable overhead efficiency variance.
17. It is caused by paying or using less than the standard amount.
19. It may not be a good outcome when buying substandard material or hiring substandard employees.
21. Causes may include substandard material, quantity discount, negotiated better price, quantity discount, or price drop.
23. Causes may include higher-quality material, better-qualified employees, or a change in manufacturing process.
25. Causes may include less-qualified employees or a change in quality level of employees due to a change in process.
27. Causes may include better material, higher-quality employees, or a change in process.

Chapter 9

Multiple Choice

1. B
3. A
5. C
7. C
9. A
11. C
13. B
15. D
17. D
19. B

Questions

1. A management control system allows management to establish, implement, and monitor the